



GeoSocial Networks

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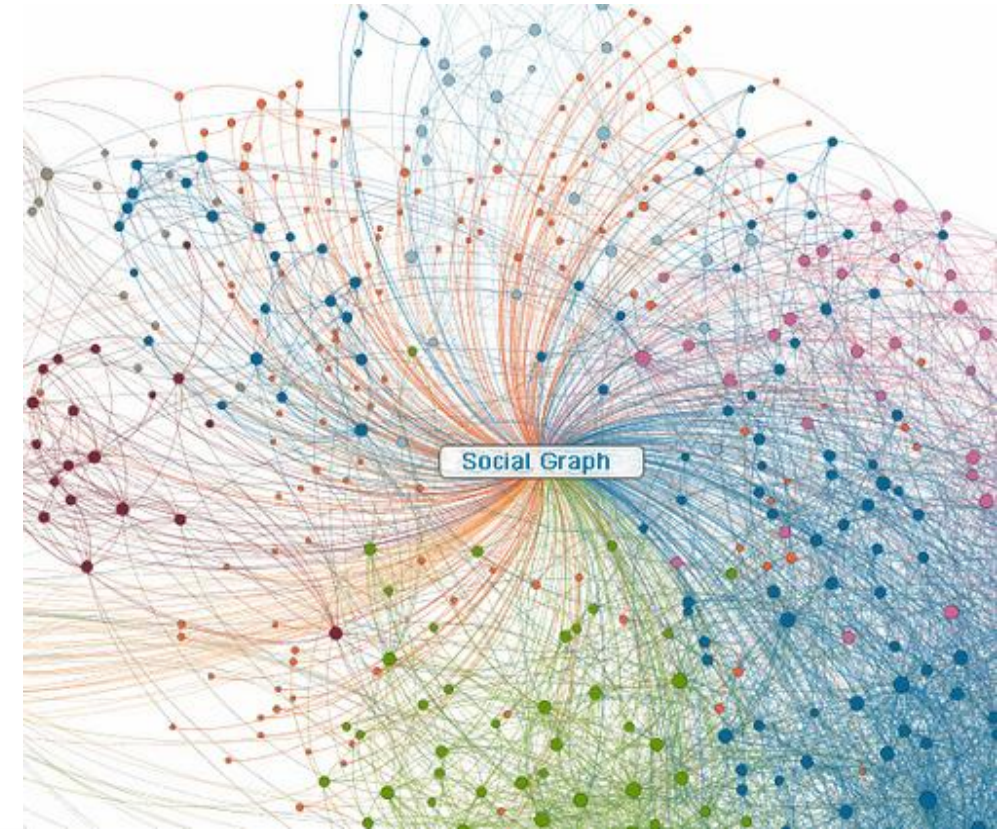
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OUTLINE



GeoSocial Queries [VLDB'13]

Inferring Social from Geo [SIGMOD'13]

GeoSocial Recommendation [SIGMOD'15]

Future [SIGSPATIAL'15]

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GeoSocial Queries [VLDB'13]

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GeoSocial Recommendation [SIGMOD'15]

Future [SIGSPATIAL'15]

Geo-Social Networks (GeoSNs)



Social network
functionality

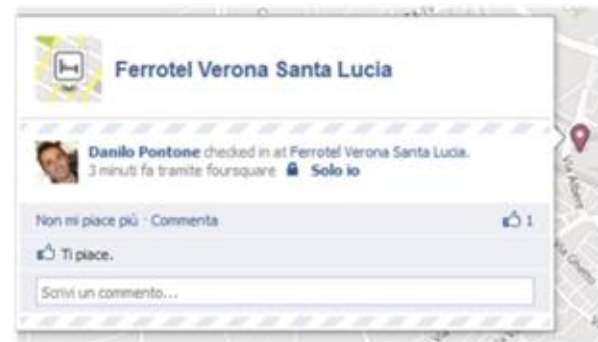
Location-based
services

Geo-Social Query

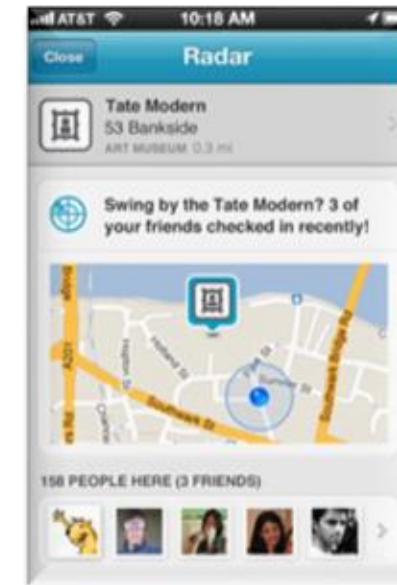
My Friends
in range



+



=



foursquare
Radar

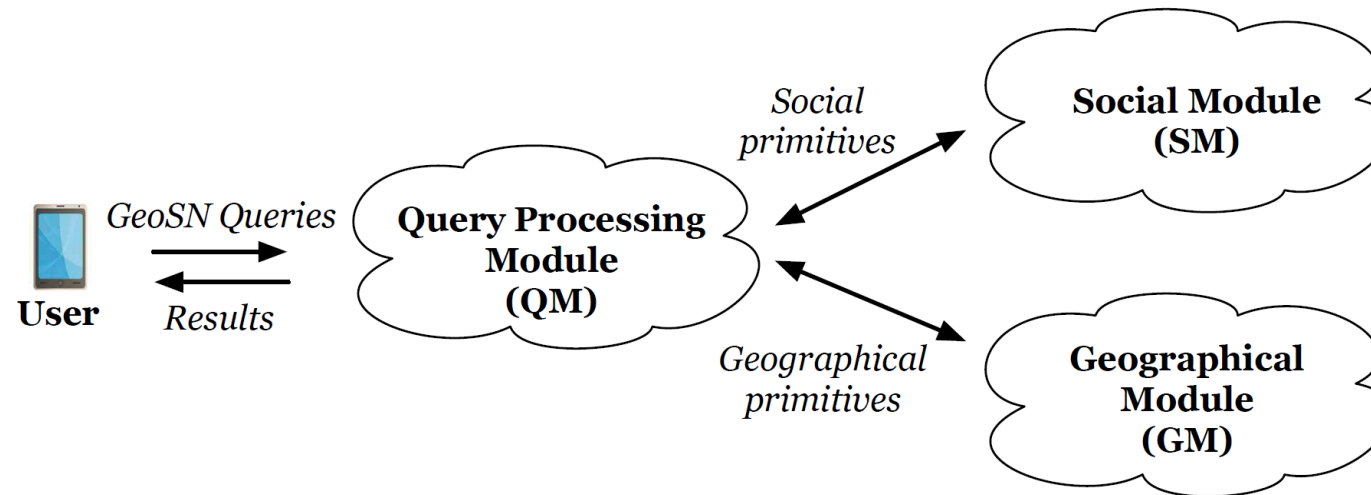


Industry & Academia

Data Management

Application/Paper	Storage Scheme
Social	
 + 	Adjacency lists in a Distributed Memory Hash Table
 + 	Adjacency lists in a Document-oriented database
[Y. Doytsher et al., WWW 2012]	Adjacency lists in Neo4j
[W. Liu et al., DASFAA 2012]	Adjacency matrix
[Y. Doytsher et al., LSBN 2010]	Edge lists in a RDBMS
Spatial	
	R*-Tree
 + 	Grids & Geohashes
[J. Bao et al., ICDE 2012]	Grid
[A. Amir et al., PMC 2007]	Quad-Tree
[W. Liu et al., DASFAA 2012]	R*-Tree

Framework Architecture



- SM and GM can be administrated by different entities.
 - Implement GeoSN queries without owning geo-social data.
- Independent functionality of social and geographical structures.
- Easy integration of new, more efficient data structures without modifications.
- Novel GeoSN query types = either a different combination of existing primitives or new ones

Framework

primitive Operations



Social Primitives

GetFriends(u)

AreFriends(u_i, u_j)

GetDegree(u)

Geographical Primitives

GetUserLocation(u)

RangeUsers(q, r)

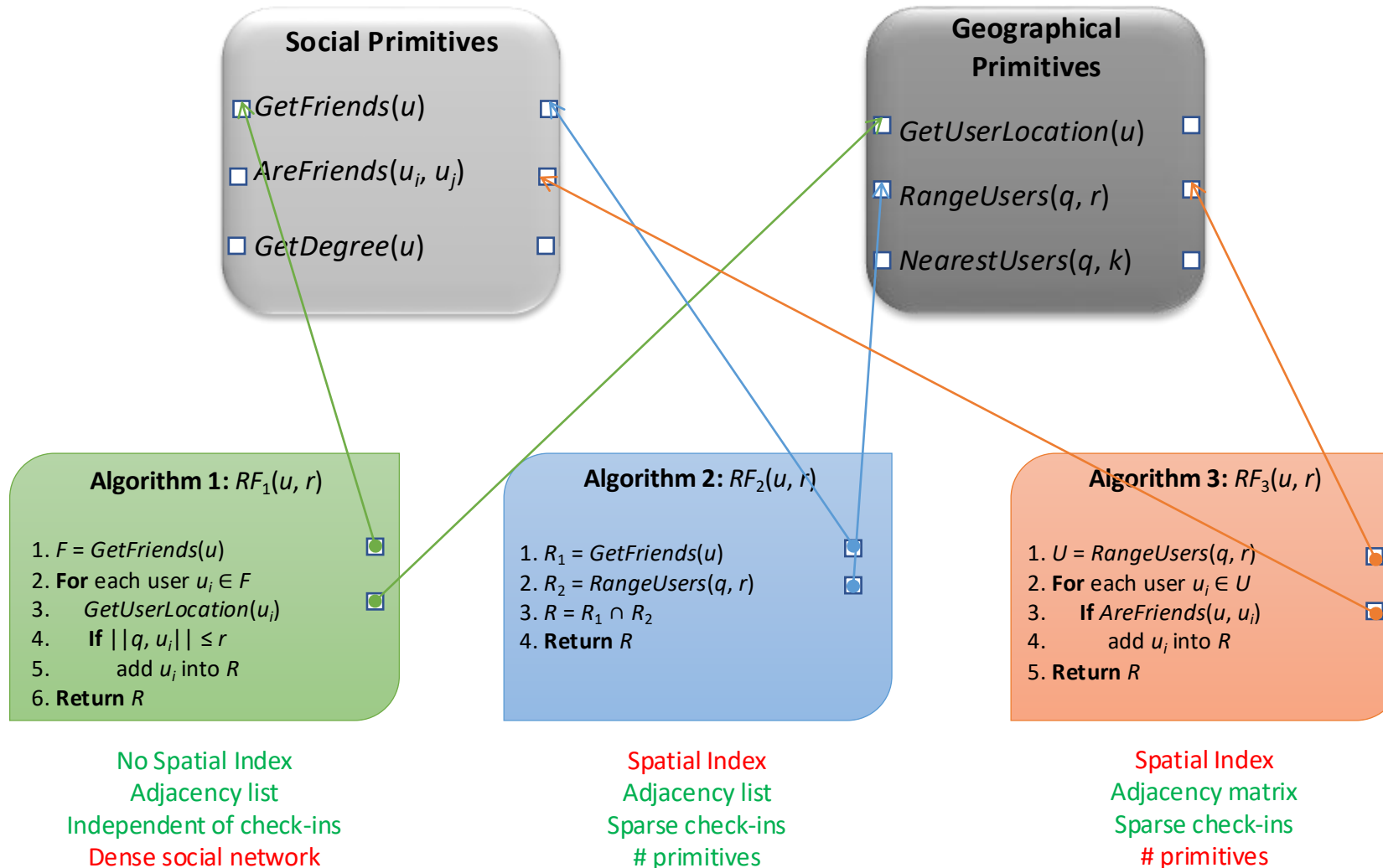
NearestUsers(q, k)

- Any primitive must be treated as an atomic operation.
 - No states.
 - *NextNearestUser* = multiple calls of *NearestUsers* – keep data locally.
 - Find more!
- Efficiency depends on the underlying storage scheme.
 - *AreFriends* - Adjacency matrix
 - *GetFriends* - Adjacency Lists
 - *GetUserLocation* – Hash Table
 - *RangeUsers* & *NearestUsers* – Spatial Indices
- They are supported by commercial GeoSNs' APIs.

Query processing

range friends

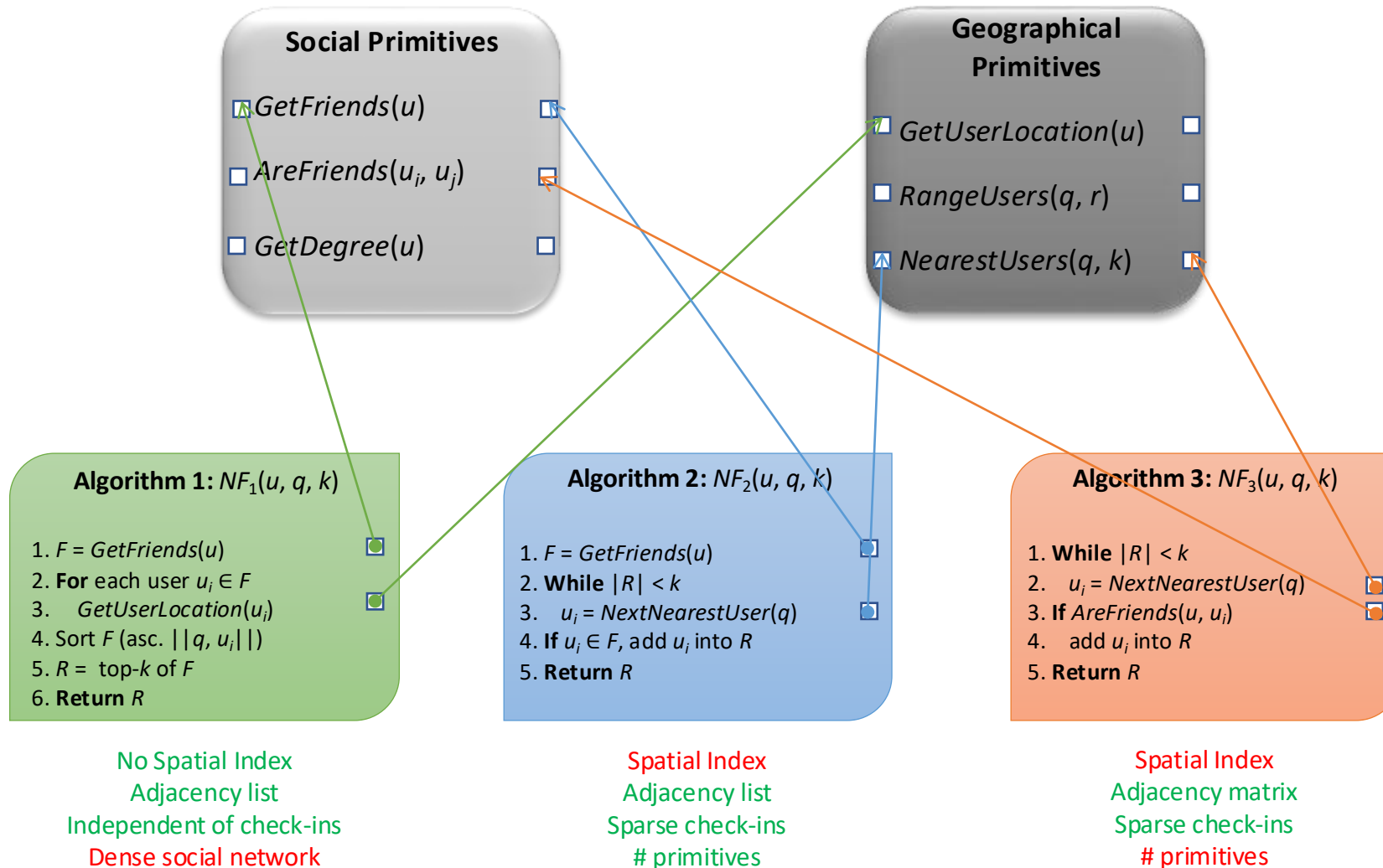
Friends of user u
within range r of q



Query processing

NEAREST FRIENDS

k nearest friends
of user u to location q .

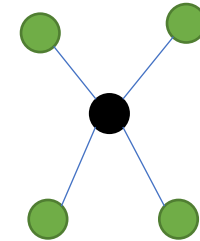


nearest star group

(NSG Query)

“the next group of five people who come to the restaurant will receive 20% discount”

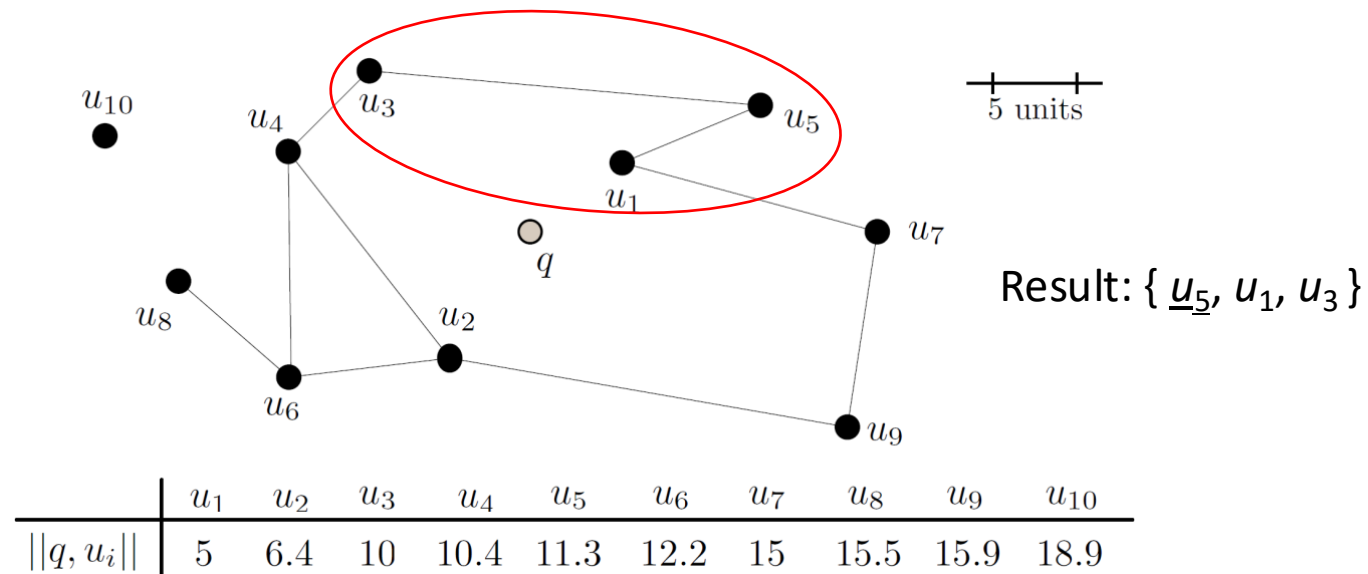
Ideally: {
 Socially connected! → Have a common friend (**star**).
 Close to the restaurant → **Min.** sum of **distances** to the restaurant



Output: k nearest groups of m users to q , such that the users in every group are connected through a common friend (star).

Nearest star group

Example ($k = 1, m = 3$)



Observation:

The **best** group of a user contains himself and his $m - 1$ closest friends to q .

NSG is not an NP-Hard problem!

Input: Location q , positive integer m

Output: Result set R

1. Initialize $b_s = \infty, b_{un} = 0, R = \emptyset, i = 1$
 2. **While** $b_{un} < b_s$
 3. $u_i = \text{NextNearestUser}(q)$
 4. $NSG_{u_i} = \{u_i\} \cup NF(u_i, q, m - 1)$
 5. **If** $|NSG_{u_i}| = m \wedge \text{adist}(q, NSG_{u_i}) < b_s$
 6. $R = (NSG_{u_i}), b_s = \text{adist}(q, NSG_{u_i})$
 7. $b_{un} = ||q, u_i|| + \sum_{(1 \leq j \leq i) \wedge (j < m)} ||q, u_j||$
 8. **If** $i < m - 1, b_{un} += (m - i - 1) \cdot ||q, u_i||$
 9. $i ++$
 10. **Return** R
-

Figure 6: Pseudocode of NSG_{eager}

NSG query processing

Basic Notation

b_s : the current lower aggregate distance achieved by the already examined users (*seen*).

b_{un} : the lower aggregate distance that can be achieved by non-retrieved users (*unseen*).

Skeleton for NSG algorithms (Branch and Bound - BnB)

Input: Location q , positive integers m, k

Output: Result set R

1. Initialize R, b_s, b_{un}
2. **While** $b_{un} < b_s$
3. Get the next nearest user to q
4. Construct his best group
5. Update result R and b_s, b_{un}
6. Refine R
7. **Return** R

Eager	Lazy	Eager [*]
Simple b_{un}	Simple b_{un}	Aggressive b_{un}
✓	✓	✓
✓	✓	✓
Find the group	Construct the graph	Find the group
✓	✓	✓
		✓
✓	✓	✓

Experiments

Linux, C++

- Storage Schemes

- Disk-based + Cache



- Social:

- **Adjacency List**: user → sorted list of friends' ids. (**document** per user)

- Geographical:

- user → coordinates (**document** per user)
 - Index: **Geohashes & Grids**

- Cache: **Linux's caching** mechanism

- Memory-based

- Social:

- (**Hash Table**) Adjacency List: user → sorted list of friends' ids.

- Geographical:

- (**Hash Table**) user → coordinates
 - Index: **Grid** (CPM)

- Machine Architecture

- Centralized: All modules at a single server.
 - Distributed: Separate server for each module (100 Mbps Ethernet)

Experiments

- Real Dataset (Foursquare & Twitter)
 - Check-ins:
 - 12,652 users
 - *same* day (May 30th, 2012)
 - in New York City (1,112 km²).
 - Social Graph:
 - 12,652 + 2M (non checked-in friends) users
 - Avg. # of friends: 437.

- Synthetic Dataset (1M, 2M, 3M, 4M, 5M)

- Check-ins
 - “The distribution of the distance between two friends follows a power law.”
 - BFS – assign locations: distance is randomly derived by the distribution in:
 - Area: 7,853 km²
- Social Graph: Barabási-Albert preference model
 - Power-law degree distribution.
 - Small-world phenomenon.
 - Avg. # of friends: 100.

[Cho et al., SIGKDD '11]

Experiments

Friends of user u within range r .

Algorithm 1: $RF_1(u, r)$

One $GetFriends(u)$
Multiple $GetUserLocation(u_i)$

Algorithm 2: $RF_2(u, r)$

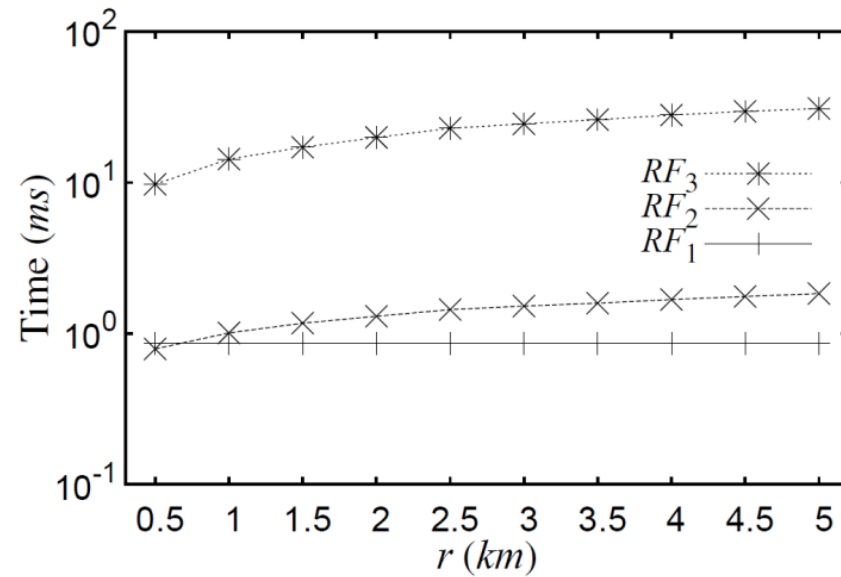
$GetFriends(u) \cap$
 $RangeUsers(q, r)$

Algorithm 3: $RF_3(u, r)$

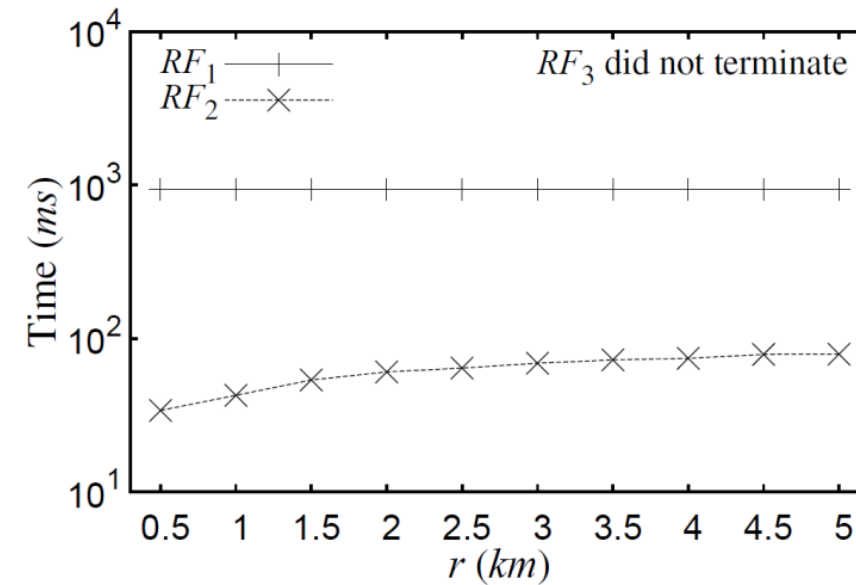
One $RangeUsers(q, r)$
Multiple $AreFriends(u, u_i)$

(Average over 100 random queries)

Real Dataset



Memory - Centralized



Memory - Distributed

Experiments

NEAREST STAR Group (NSG)

NSG_{eager}

For each newly retrieved user compute his best group eagerly.

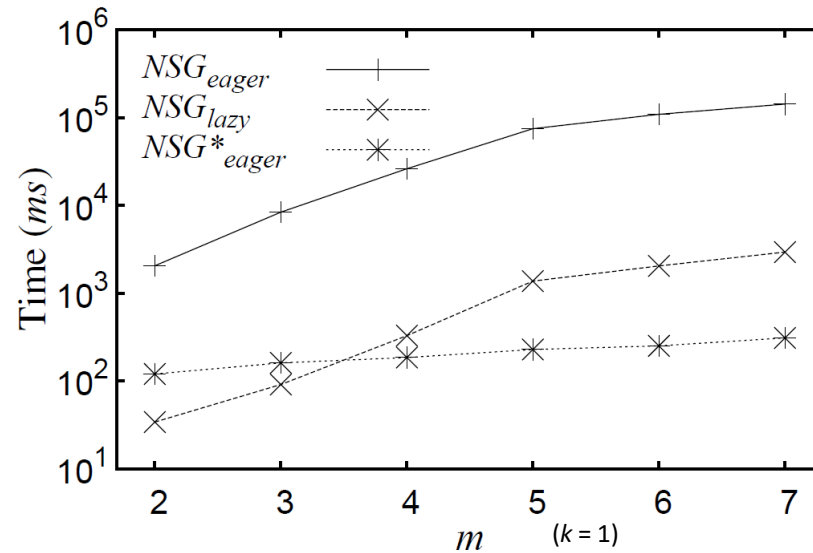
NSG_{lazy}

Construct the social graph around q iteratively.

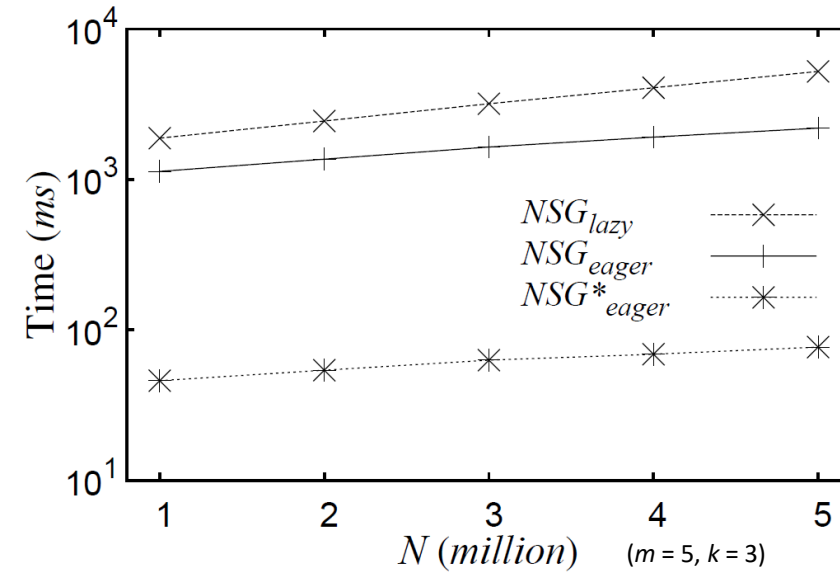
NSG^*_{eager}

Similar to NSG_{eager} , but more aggressive bounds. Refinement step.

(Average over 100 random queries + warm up)



Real Dataset : Disk - Centralized



Synthetic Dataset : Memory - Centralized

- In the most of the cases NSG^*_{eager} is the best.
- Performance scales well with the dataset size.

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Location-Enriched Datasets



- Popularity of Location-Based Services





Location-Enriched Datasets

- Popularity of Location-Based Services

Twitter: 10M+ geo-tagged tweets/day mashable.com

Foursquare: 5M check-ins/day venturebeat.com/2015/08/09/



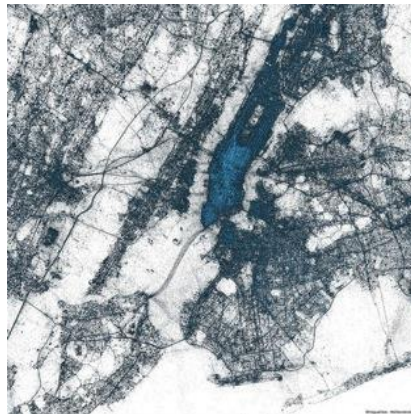


Location-Enriched Datasets

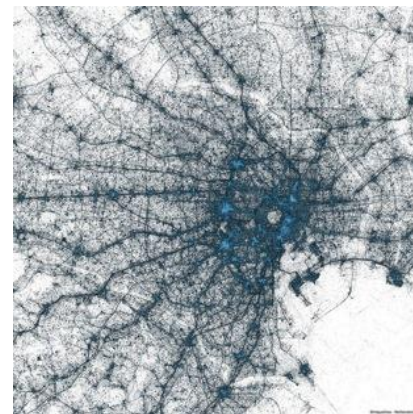
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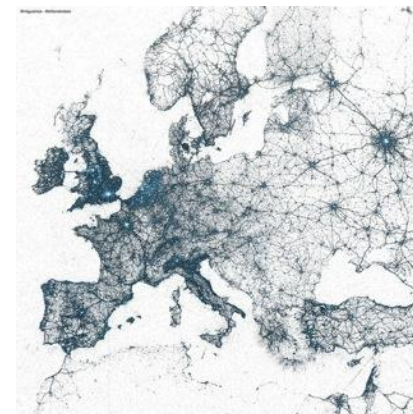
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New York City



Tokyo



Europe

Geo-Tagged Tweets on
Map

by Twitter mashable.com

Social Relationship Inference from Location Data



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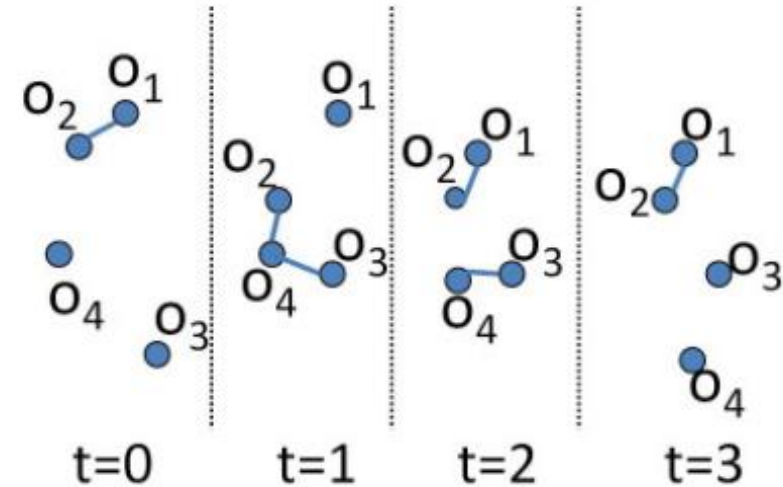


- Reachability [VLDB'12]
 - u is reachable to v in time period T
 - if there is a **contact path**

Social Relationship Inference from Location Data



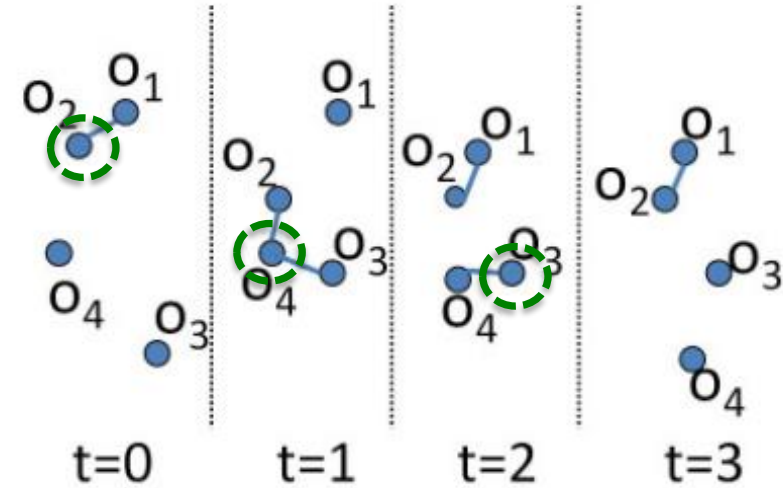
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Path: $O_1 \rightarrow O_2 \rightarrow O_4 \rightarrow O_3$ between time $[0,2]$

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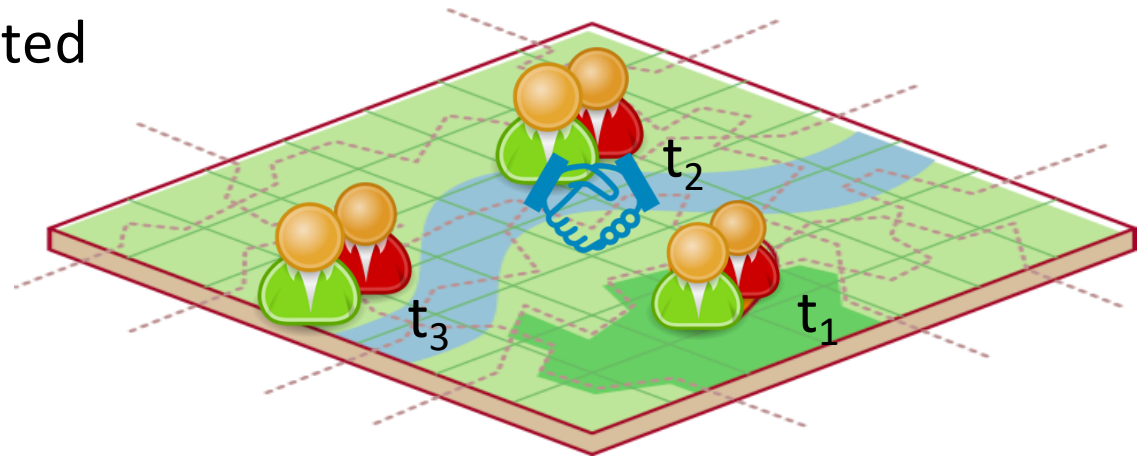


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- Social Strength [SIGMOD'13]
 - u and v are socially connected
 - how often they **meet** and **where**

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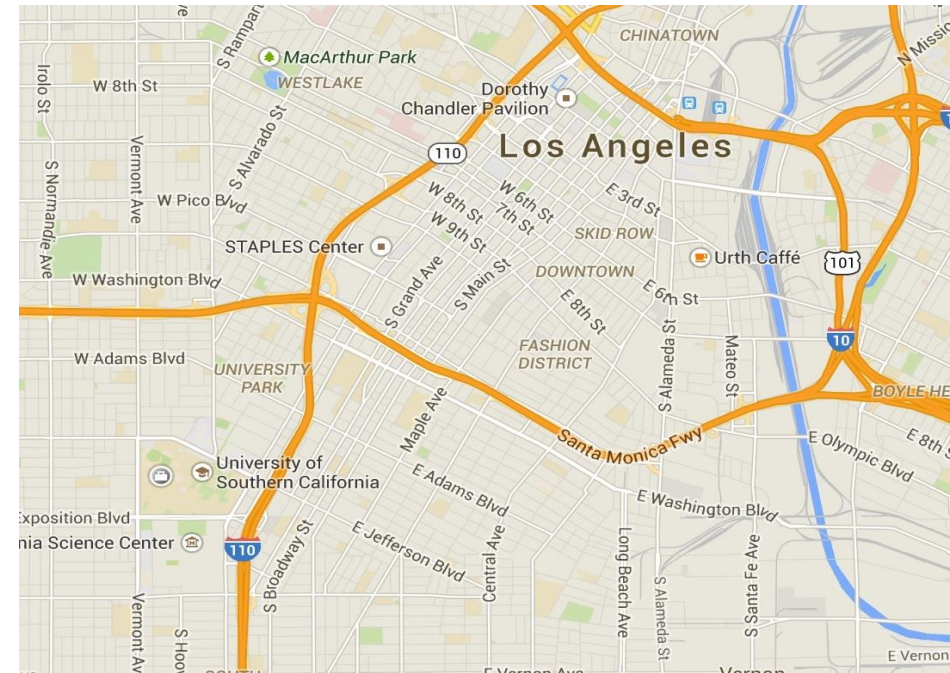


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 - u influences v
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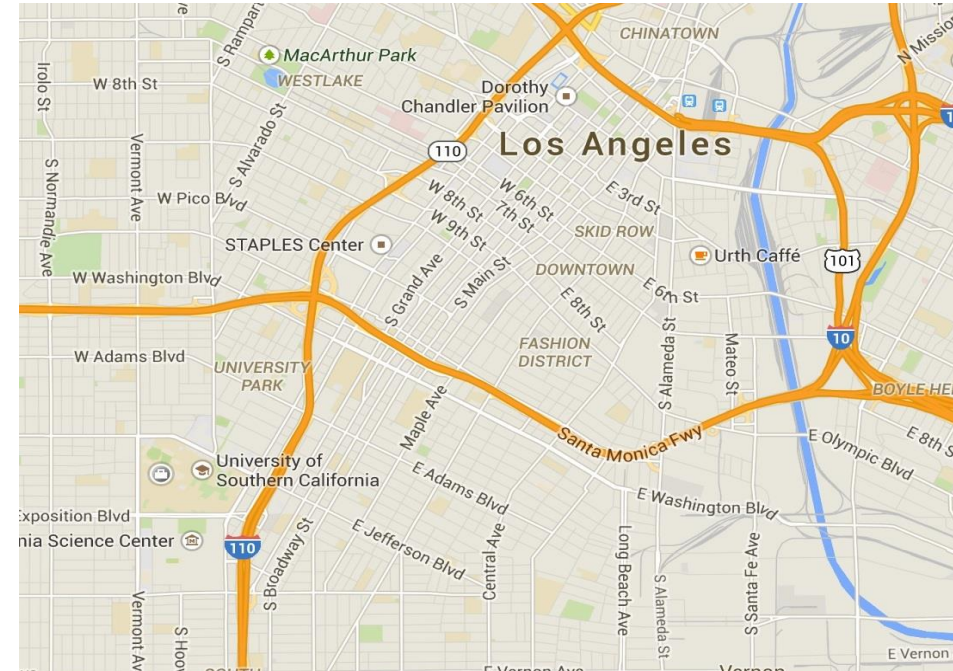
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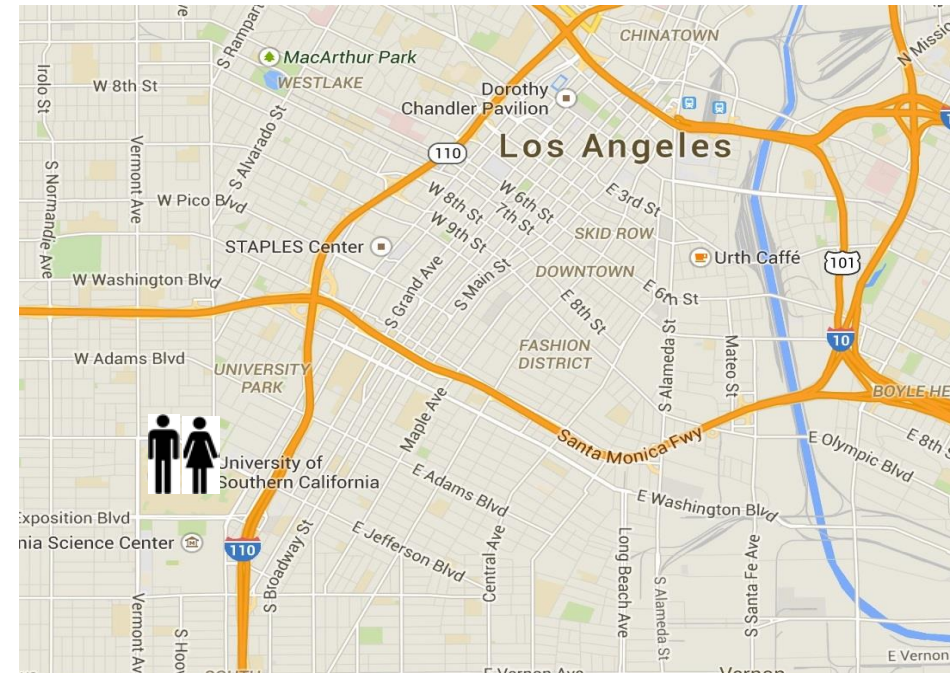


followship: u visits a place, and v visits the same place shortly after.

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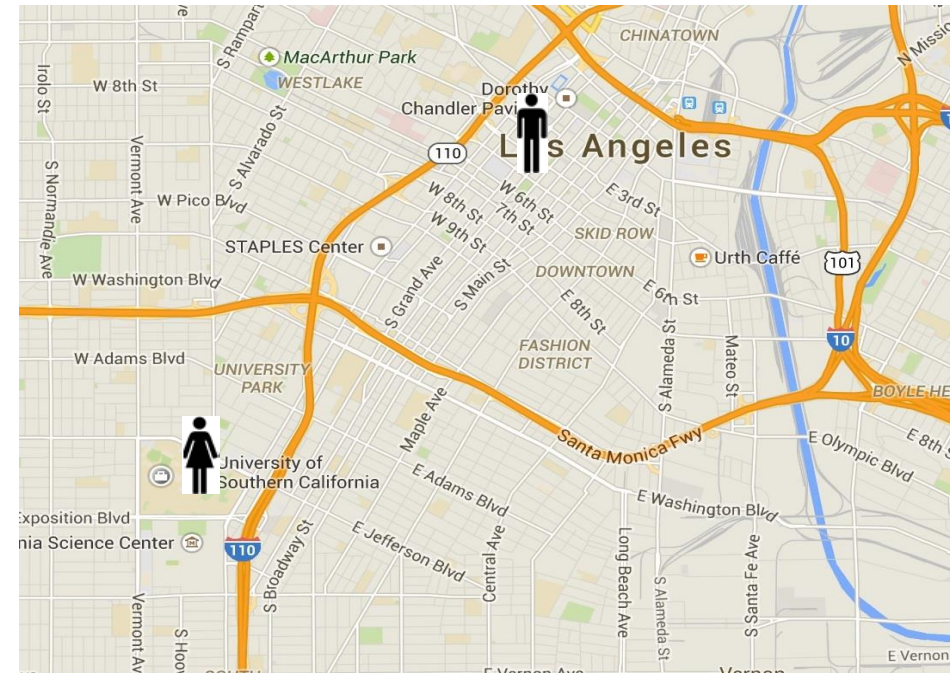


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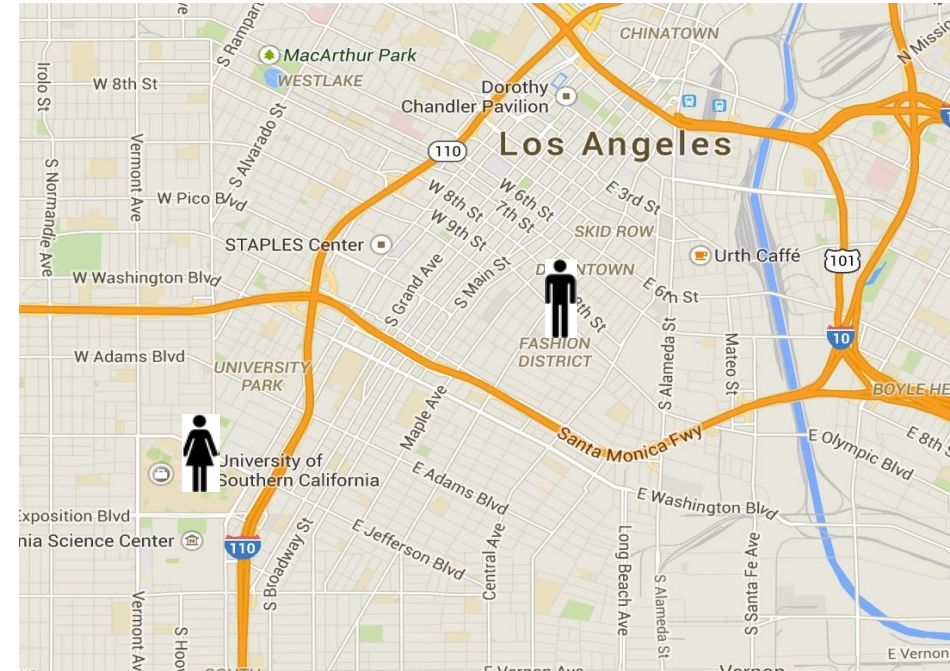


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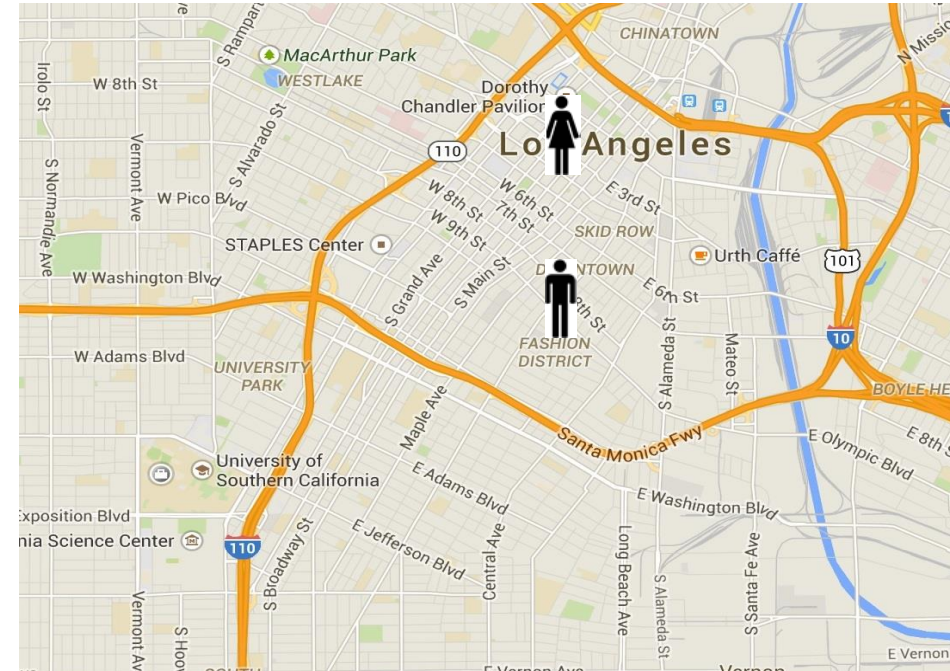


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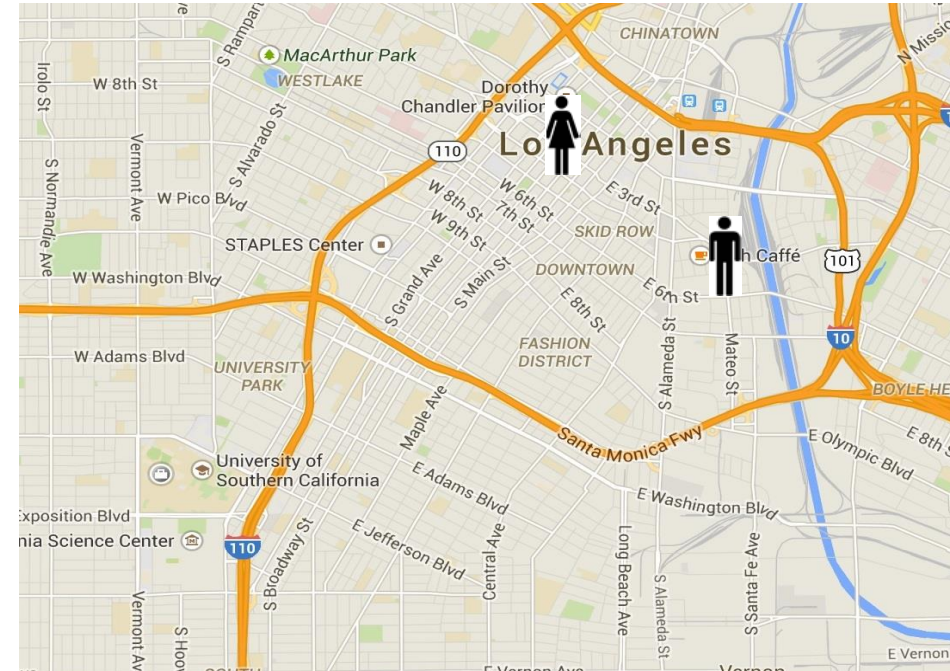


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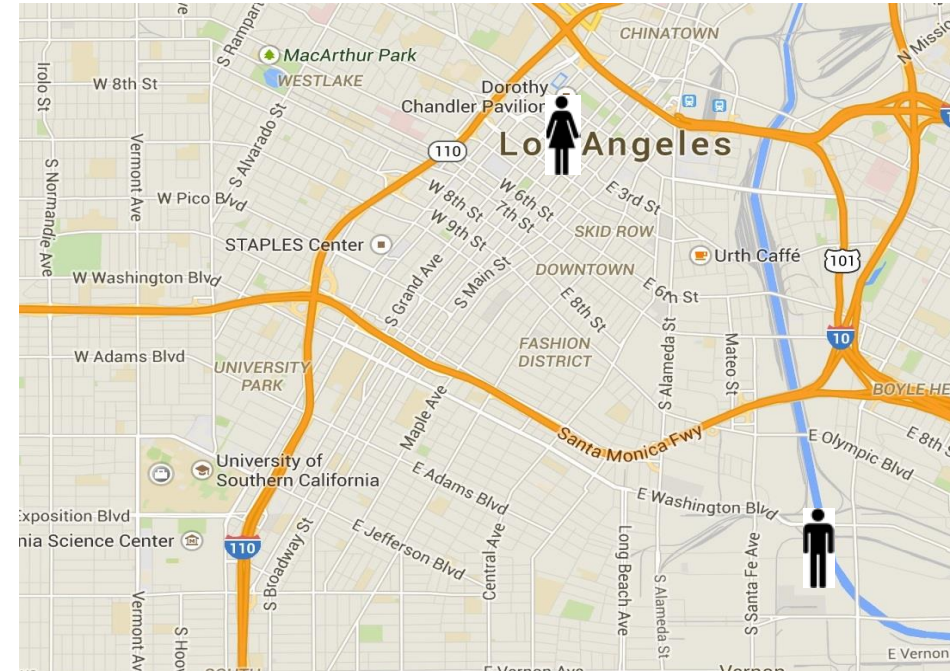


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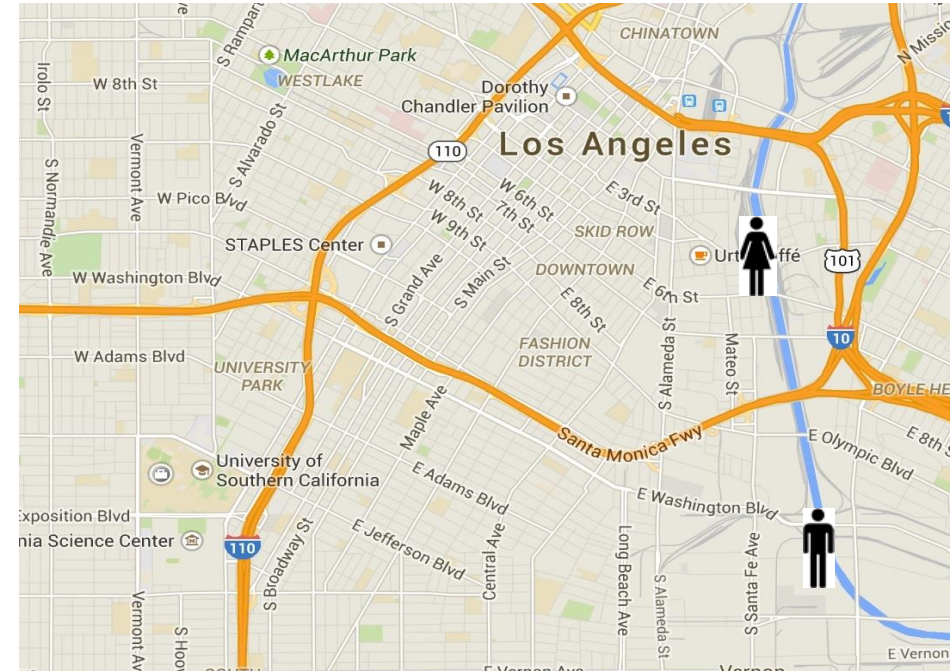


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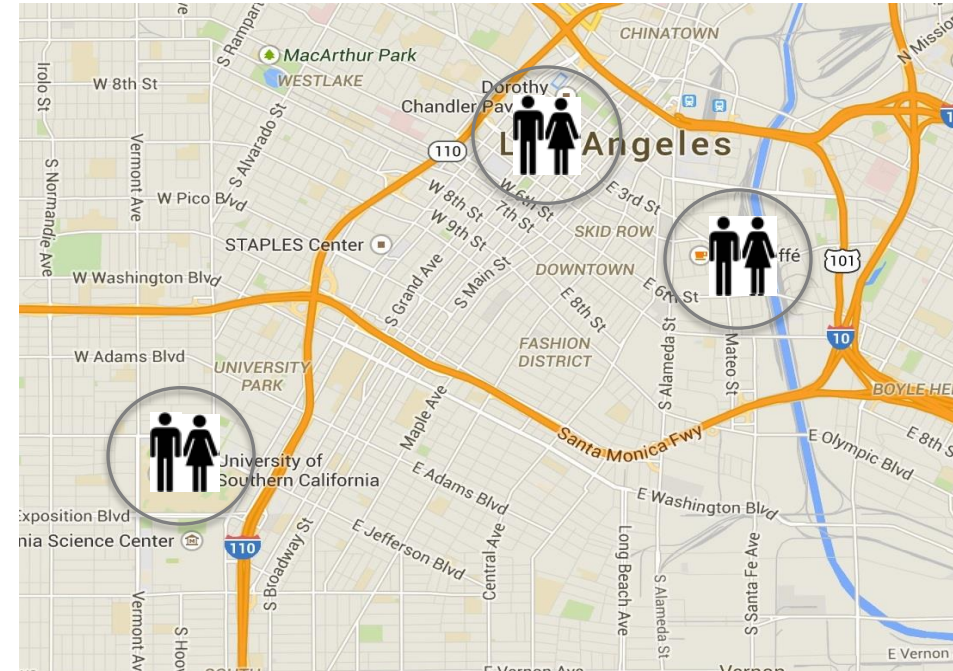


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Applications



Applications



- Social Network
 - Marketing
 - Friendship suggestions
 - Social and cultural studies

Applications

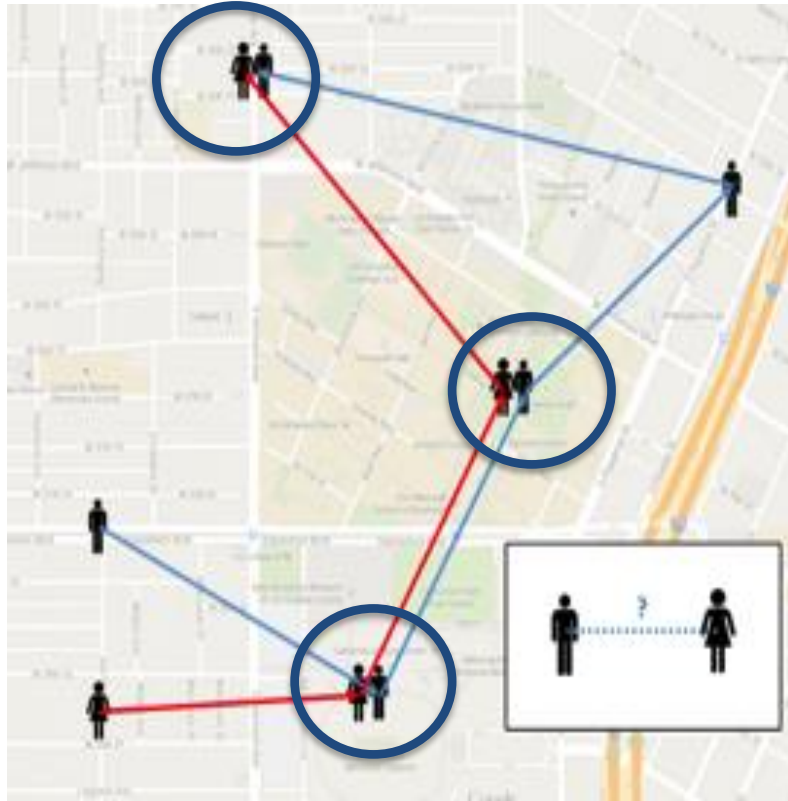


- Social Network
 - Marketing
 - Friendship suggestions
 - Social and cultural studies
- Geo-social Network
 - Criminology
 - identify the new or unknown members of a criminal gang or a terrorist cell
 - Epidemiology
 - spread of diseases through human contacts
 - Policy
 - induce local influence in electing a tribal representative

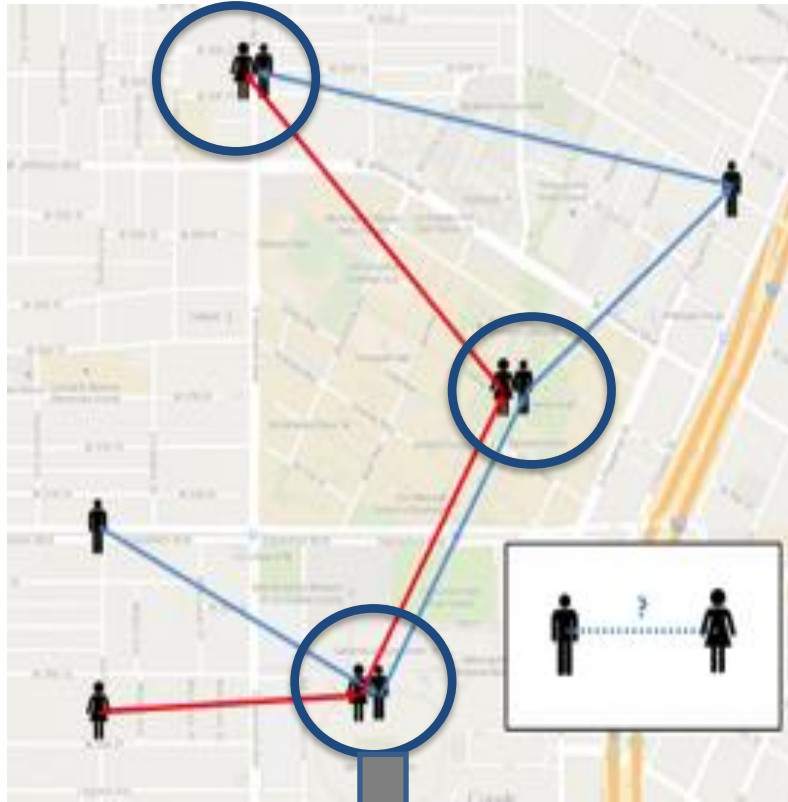


Real-World Social Strength - Intuition



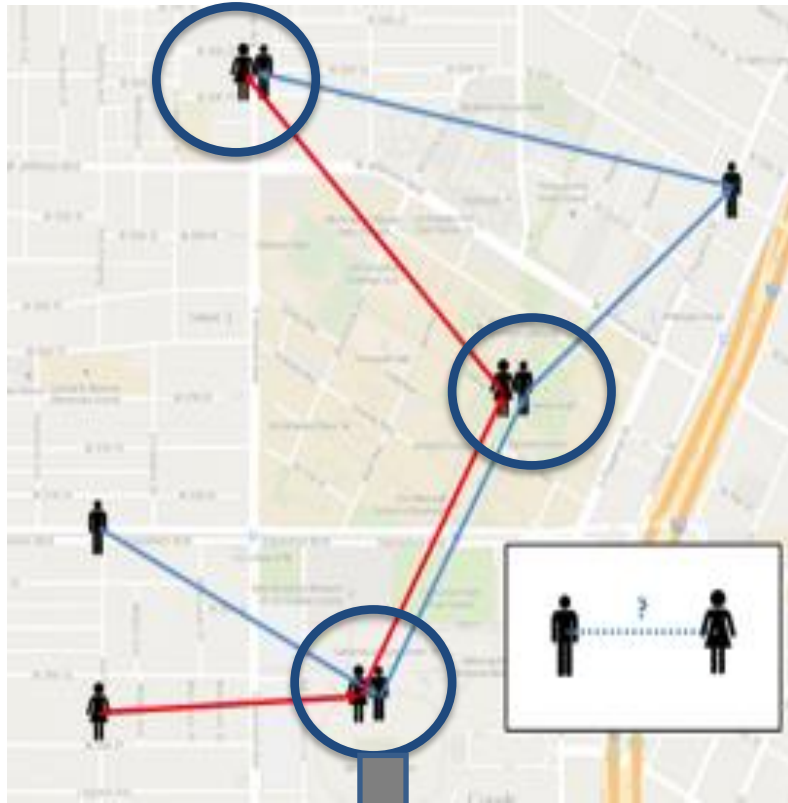


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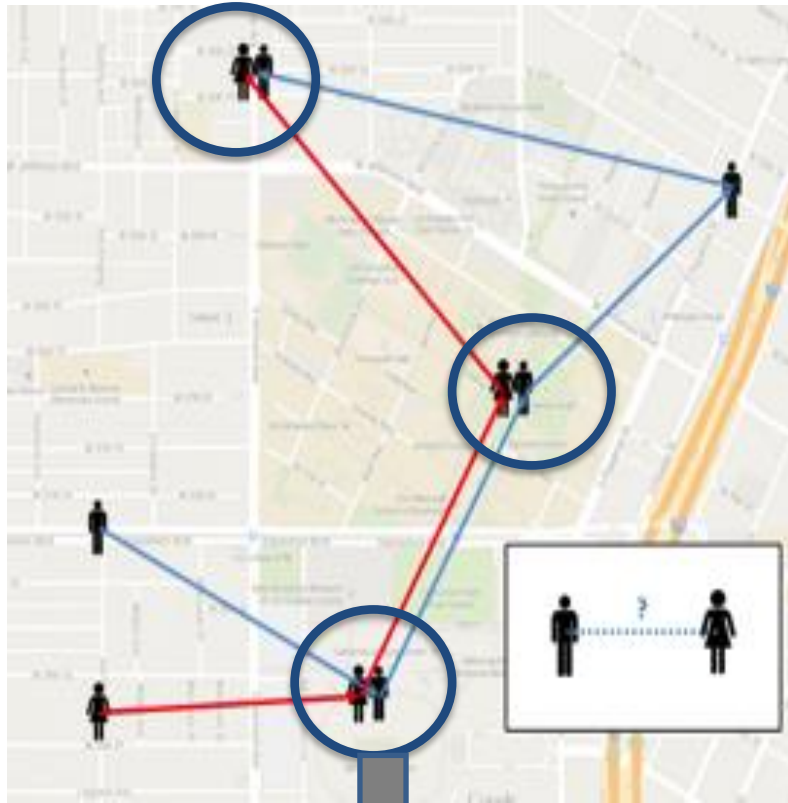
Co-occurrence

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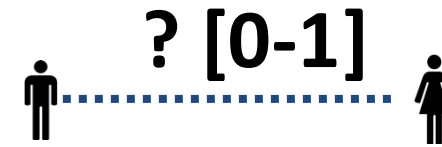


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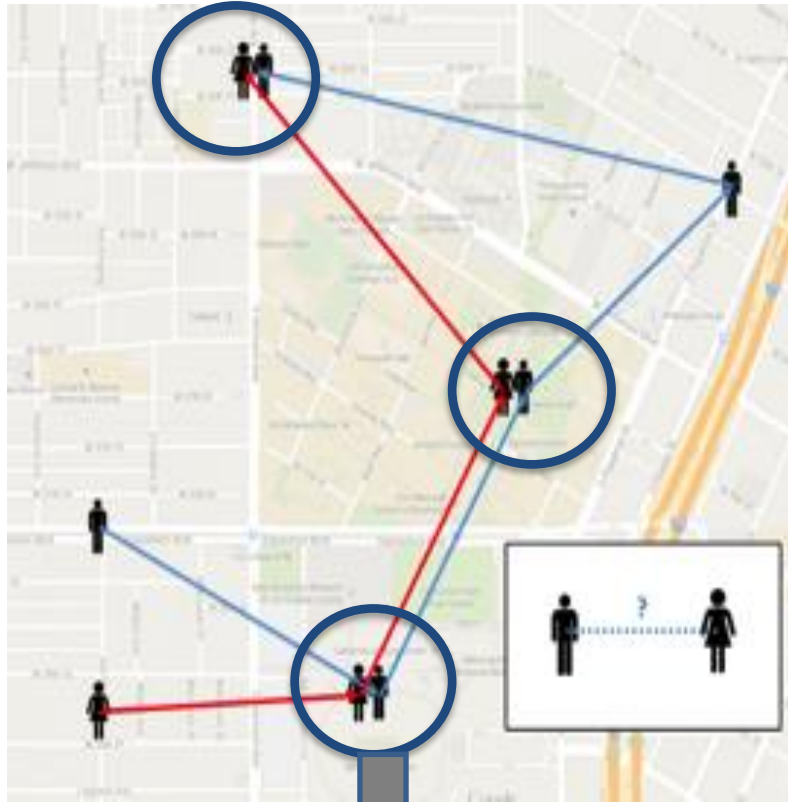


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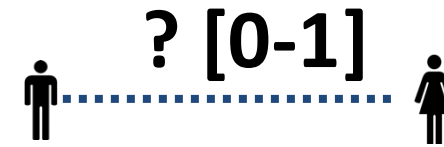




Real-World Social Strength - Intuition



Co-occurrence



From Real-World *Co-occurrences*
to *Social Strength*

Inferring friendship network structure by using mobile phone data (PNAS'09)

N. Eagle, A. Pentland, D. Lazer



- ❖ Study traces of 94 subjects using mobile phones
 - Subjects also reported their data: proximity and friendships
 - Analyzes proximity and friendships (inferred from recorded data) vs. ones that were self-reported by users
 - Conc-1: Two data sources is overlapping but distinct
 - Conc-2: Accurately infer 95% of friendships based on the observational data alone, where friend dyads demonstrate distinctive temporal and spatial patterns in their physical proximity and calling patterns.

Inferring social ties from geographic coincidences (in PNAS'10)



David J. Crandall, Lars Backstromb, Dan Cosleyc, Siddharth Surib,
Daniel Huttenlocher, and Jon Kleinberg

❖ Probabilistic Model

- Infer the probability of two people being friends given their co-occurrences in space and time
- Does not consider the frequency of co-visit
- Simplifies the social network: one connection for each person

Bridging the Gap between Physical Location and Online Social Network (UbiComp '10)



J. Cranshaw, E. Toch, J. Hong, A. Kittur, N. Sadeh

- Introduces a novel set of location based features for analyzing the social context of a geographical region
- **Location Entropy**: analyzes **the** context of the social interactions at that location: crowdedness and diversity
- **Regularity (Schedule_Entropy)**: High value reflects irregular movements, which produce high chance of making new friends
- Establishes a model of friendship in an online social network based on contextual features of co-locations

Example



Example

USC



SM Pier



Kodak Theater



Example

USC



SM Pier



Kodak Theater



(u_1, u_2)

(u_2, u_3)

(u_1, u_3)

Example

USC



SM Pier



Kodak Theater



(u_1, u_2)

4

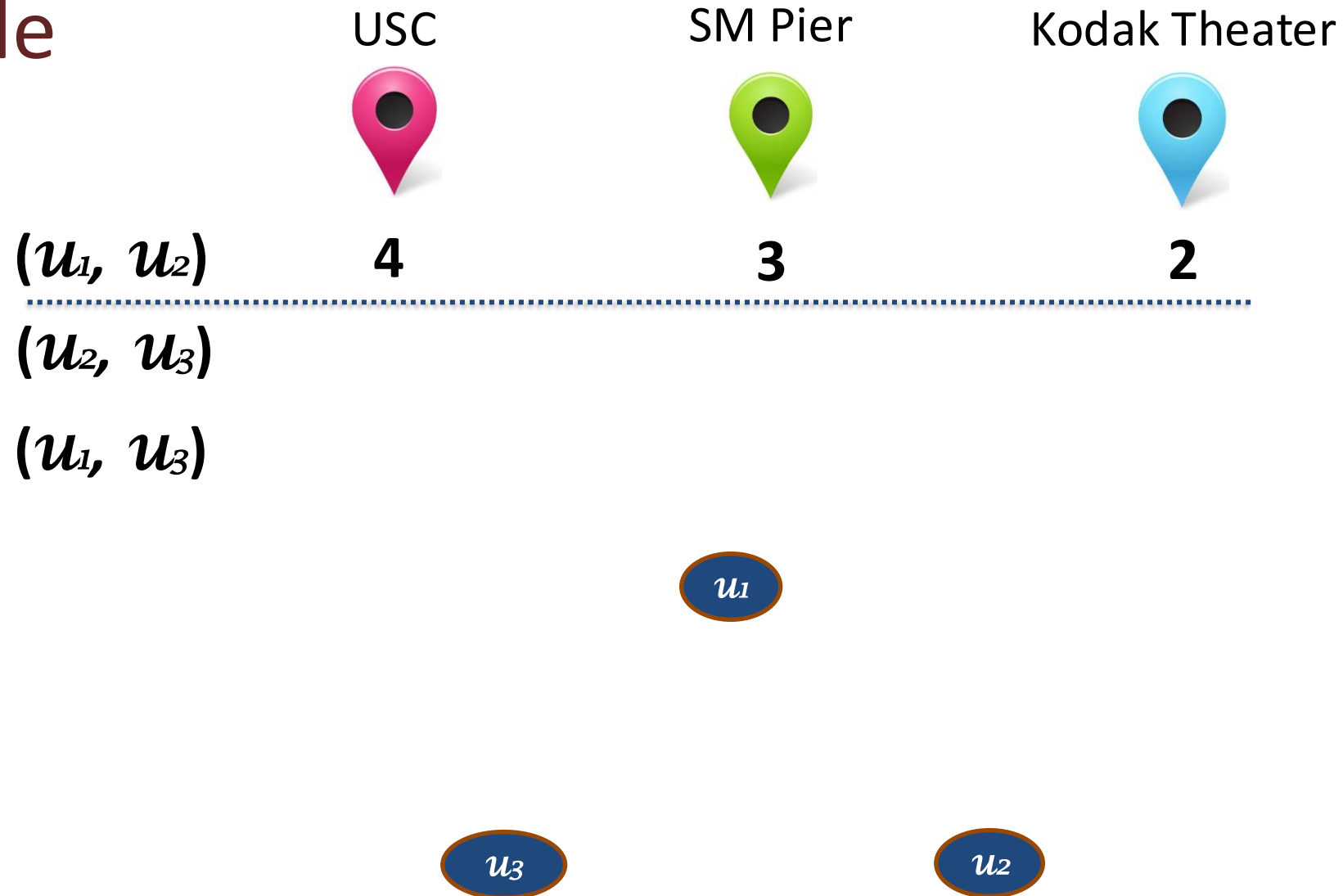
3

2

(u_2, u_3)

(u_1, u_3)

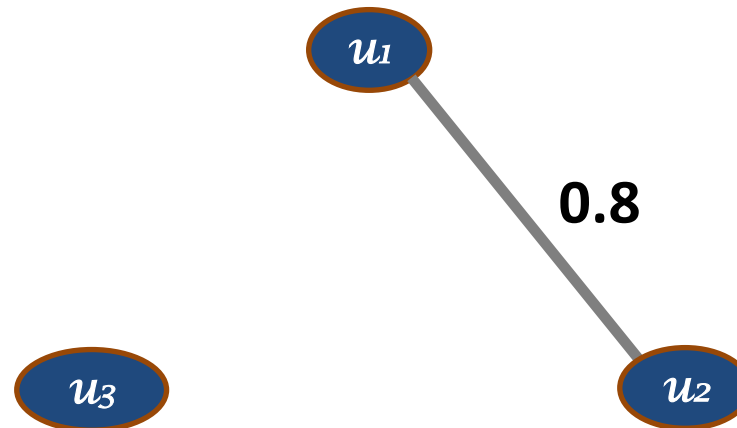
Example



Example



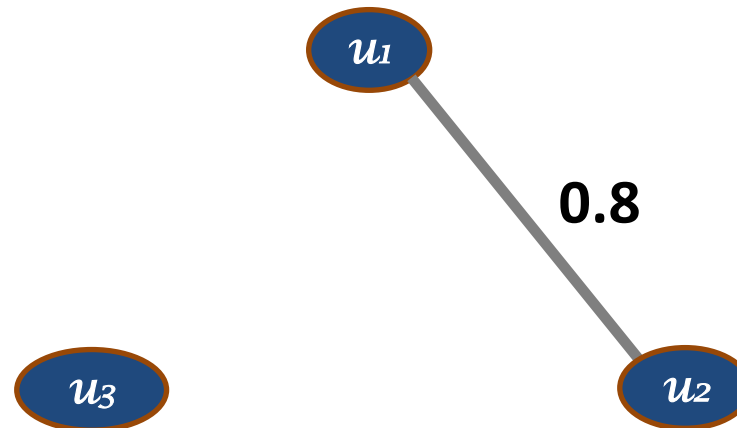
	USC	SM Pier	Kodak Theater
(u_1, u_2)	4	3	2
(u_2, u_3)			
(u_1, u_3)			



Example



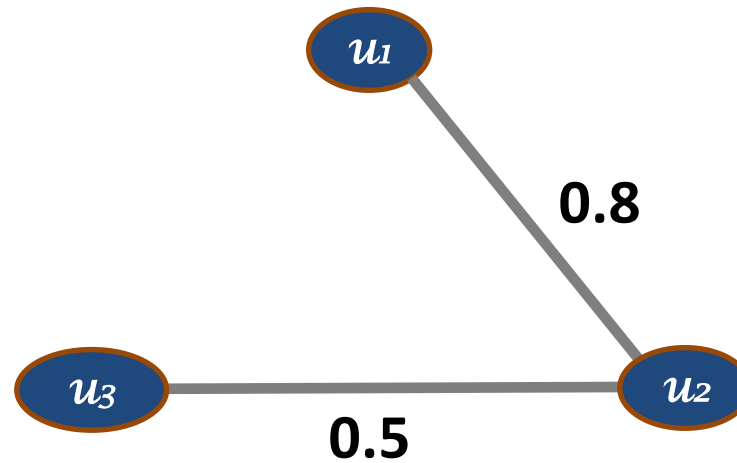
	USC 	SM Pier 	Kodak Theater 
(u_1, u_2)	4	3	2
(u_2, u_3)	2	2	1
(u_1, u_3)			



Example



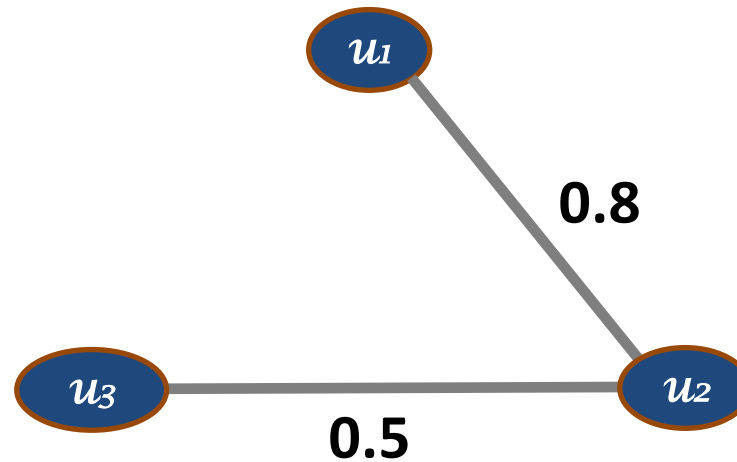
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Example



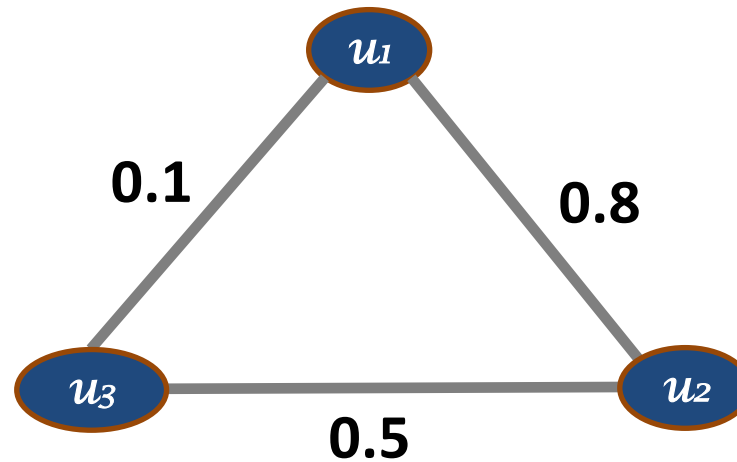
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(u_1, u_3)		5	



Example



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Social strength is a quantitative measure that tells how socially close two people are.



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Input: Users: $U = (u_1, u_2, \dots, u_M)$ Locations : $L = (l_1, l_2, \dots, l_N)$

Spatiotemporal records $\langle user_id, location, time \rangle : \langle u, l, t \rangle$



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Spatiotemporal records $\langle user_id, location, time \rangle : \langle u, l, t \rangle$

Output: a weighted social graph where the weights of the edges define social strengths.

Challenges





Challenges

1. What features of co-occurrences matter?
 - Richness?
 - Frequency?
 - Coincidences?



Challenges

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 - Richness?
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2. Location
 - Popularity?
 - Semantics?



Challenges

1. What features of co-occurrences matter?
 - Richness?
 - Frequency?
 - Coincidences?
2. Location
 - Popularity?
 - Semantics?
3. Quantify friendships
 - Social Strength in between $[0,1]$

Baseline Solution - Richness





Baseline Solution - Richness

Counting the number of unique locations



Baseline Solution - Richness

Counting the number of unique locations

Co-occurrence Vectors

$$C_{12} = (10, 1, 0, 0, 9)$$

$$C_{23} = (2, 3, 2, 2, 3)$$

$$C_{13} = (10, 0, 0, 0, 10)$$



Baseline Solution - Richness

Counting the number of unique locations

Co-occurrence Vectors

Richness

$$C_{12} = (10, 1, 0, 0, 9)$$

3

$$C_{23} = (2, 3, 2, 2, 3)$$

5

$$C_{13} = (10, 0, 0, 0, 10)$$

2



Baseline Solution - Richness

Counting the number of unique locations

Co-occurrence Vectors	Richness
$C_{12} = (10, 1, 0, 0, 9)$	3
$C_{23} = (2, 3, 2, 2, 3)$	5
$C_{13} = (10, 0, 0, 0, 10)$	2

✗ Ignore multiple co-occurrences @ same places

Baseline Solution - Frequency





Baseline Solution - Frequency

Counting the number of co-occurrences



Baseline Solution - Frequency

Counting the number of co-occurrences

Co-occurrence vectors

$$C_{13} = (10, 1, 0, 0, 9)$$

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Baseline Solution - Frequency

Counting the number of co-occurrences

Co-occurrence vectors

Frequency

$$C_{13} = (10, 1, 0, 0, 9)$$

20

$$C_{23} = (2, 3, 2, 2, 3)$$

13

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Baseline Solution - Frequency

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$C_{31} = (10, 0, 0, 0, 10)$	20

- ✓ Captures local frequency
- ✗ Cannot capture the diversity of co-occurrences

EBM Model [SIGMOD'13]

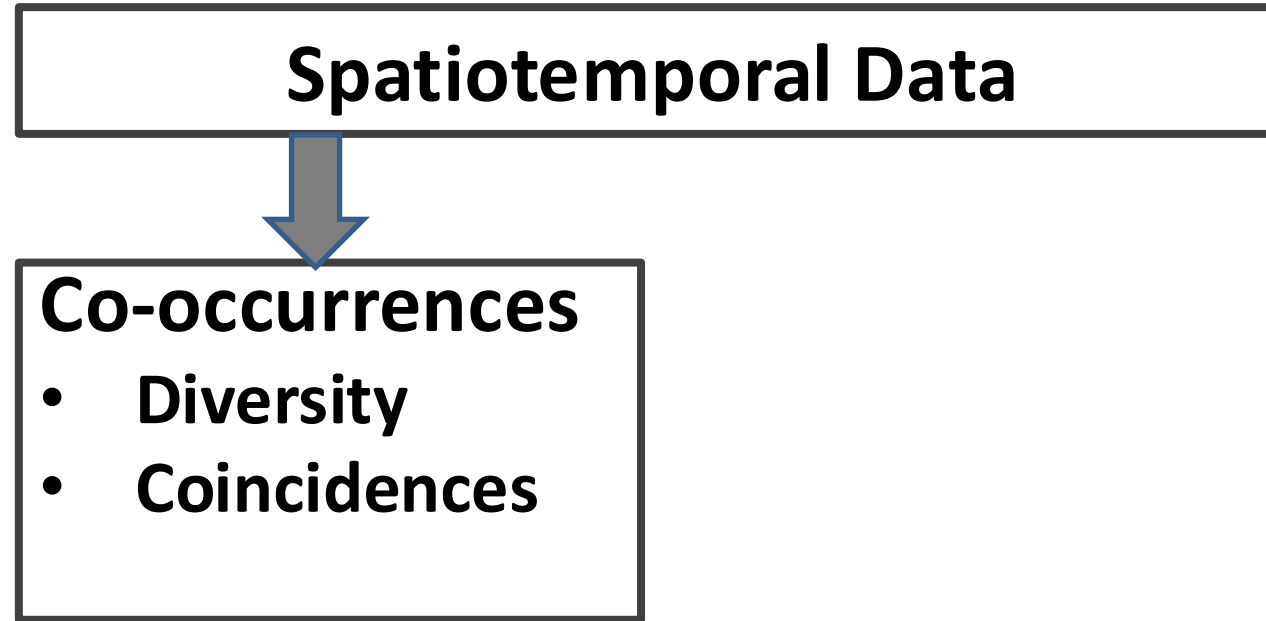


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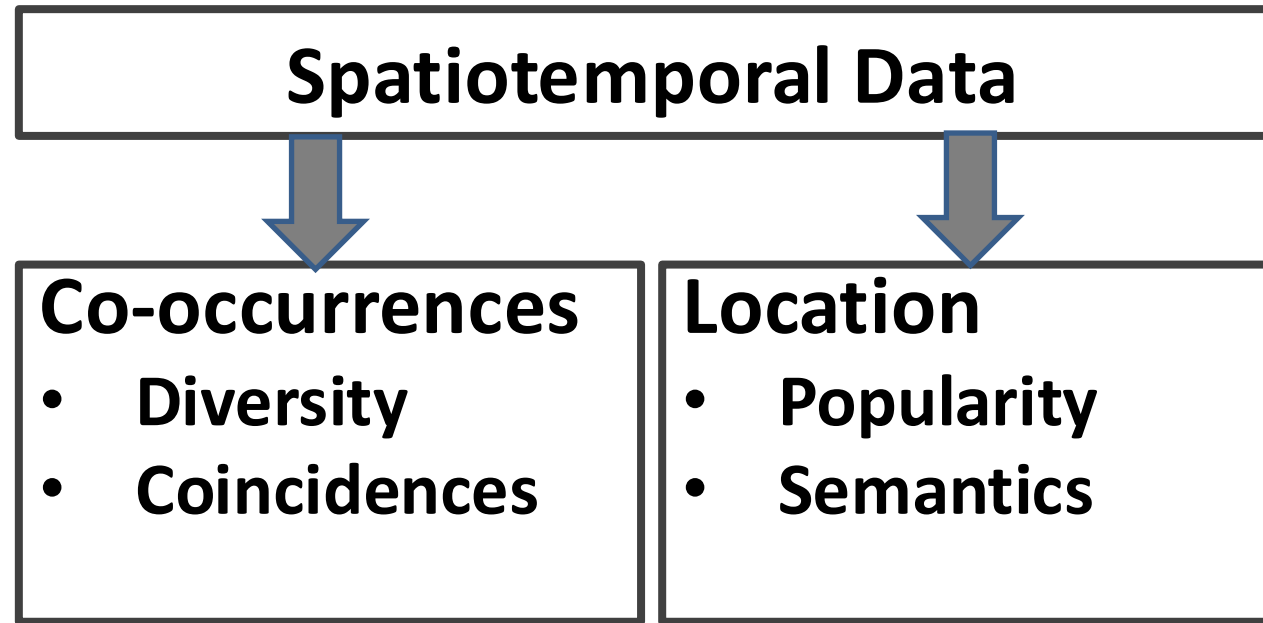


Spatiotemporal Data

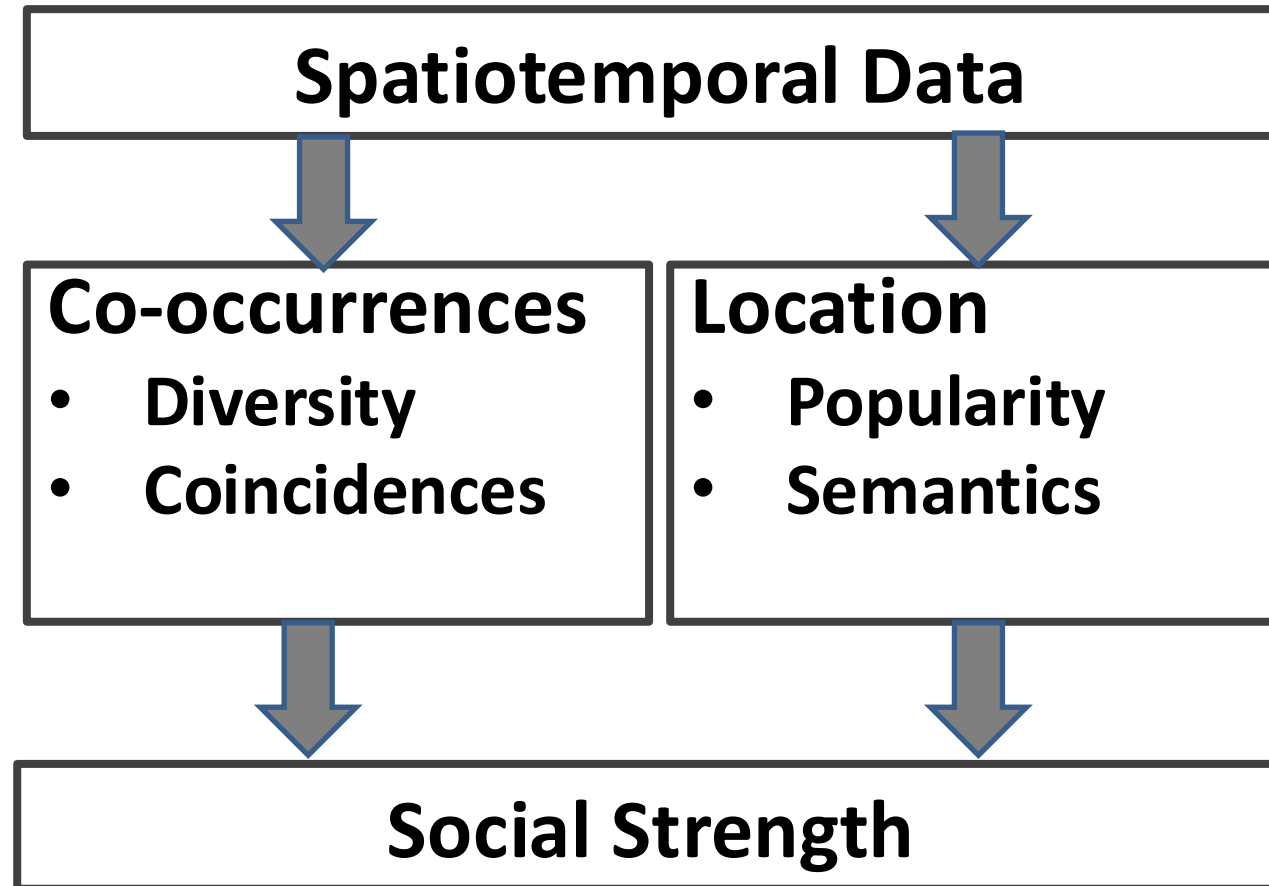
EBM Model [SIGMOD'13]



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EBM Model [SIGMOD'13]



Shannon Entropy



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- More diverse places they co-occurred $\rightarrow$ Low predictability $\rightarrow$ High entropy



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H_{ij}^S

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1.59

0.69



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✓ The more locations, the higher entropy.



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- ✓ The more locations, the higher entropy.
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✓ The more locations, the higher entropy.

✓ The more diverse, the higher entropy.

✗ No control on diversity vs. frequency, e.g., may put too much weight on outliers (coincidences)

Rényi Entropy



Rényi Entropy

We want to control the impact of diversity vs. frequency



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We want to control the impact of diversity vs. frequency

$$H_{ij}^R = \left(-\log \sum_l \left(P_{ij}^l \right)^q \right) / (q - 1)$$



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Order of diversity

Two blue arrows point from the text "Order of diversity" to the exponent 'q' in the equation and the denominator '(q - 1)', indicating that the parameter q controls the order of diversity.



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$$H_{ij}^R = \left(-\log \sum_l \left(P_{ij}^l \right)^q \right) / (q - 1)$$

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- $q = 1$ – Renyi entropy is *undefined*, but its limit exists and becomes **Shannon** entropy, where it is unbiased.



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- $q = 1$ – Renyi entropy is *undefined*, but its limit exists and becomes **Shannon** entropy, where it is unbiased.
- $q = 0$ the entropy is *insensitive* to local frequencies $\Leftrightarrow$ giving pure number of unique locations – **richness**.

Rényi Entropy



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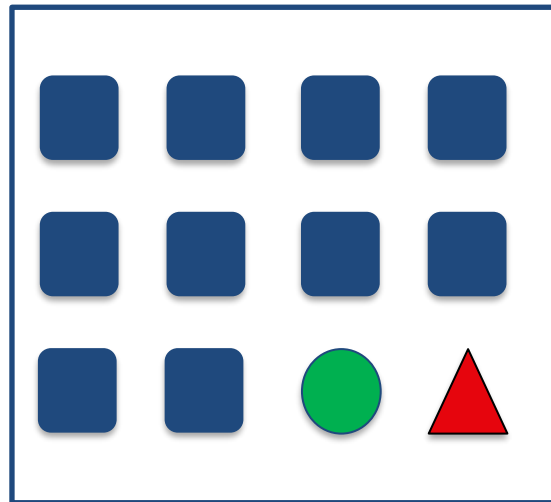
- ✓ Captures the diversity of co-occurrences.
- ✓ Limits impact of coincidences (outliers).
- ✗ Still considers all locations equally important. We need to consider:
 - Location popularity

Location Entropy for Location Popularity

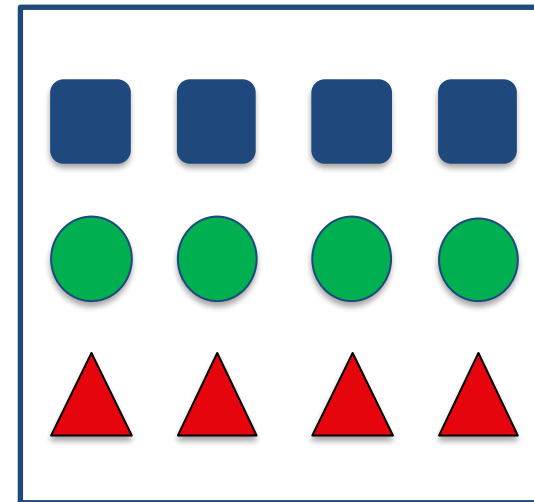


Frequency = 12

Diversity = 3



Location 1



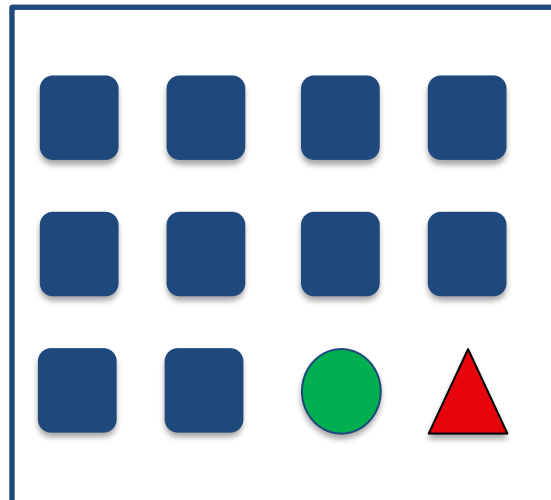
Location 2



Location Entropy for Location Popularity

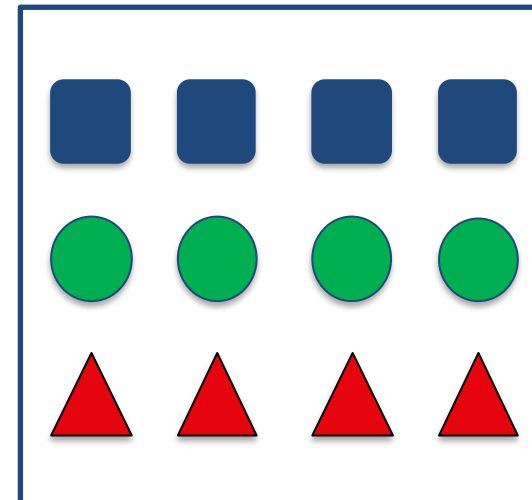
Frequency = 12
Diversity = 3

Less
Popular
 $LE = 0.566$



Location 1

More
Popular
 $LE = 1.099$



Location 2



Location Entropy (LE)

$$H_l = - \sum_{u, P_{u,l} \neq 0} P_{u,l} \log P_{u,l}$$

- LE indicates the popularity of a location *Cranshaw, J., et al., (2010). Bridging the gap between physical locations and online social networks. UBICOMP, 119-128.*
- The more popular, the higher entropy, and vice versa
- LE captures how diverse the visitors of a location are
 - E.g., your home is not diverse as only 2-4 users visited there; Eifel tower is the opposite
- Pick a random visit v at location l ; high entropy means:
 - less predictable who made v
 - *The location has more diverse set of visitors*

The Entropy Based Model (EBM)





The Entropy Based Model (EBM)

- Renyi Entropy

$$H_{ij}^R = \left(-\log \sum_l \left(P_{ij}^l \right)^q \right) / (q - 1)$$

(How often i and j meet in how diverse of locations)



The Entropy Based Model (EBM)

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(How often i and j meet in how diverse of locations)

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(How popular a location is)

- Weighted Frequency

$$F_{ij} = \sum_l c_{ij,l} \times \exp(-H_l)$$

(More weights to meetings in unpopular locations)



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(How popular a location is)

- Weighted Frequency

$$F_{ij} = \sum_l c_{ij,l} \times \exp(-H_l)$$

(More weights to meetings in unpopular locations)

- Social Strength

$$s_{ij} = \alpha. \exp(H_{ij}^R) + \beta. \sum c_{ij}^l \times \exp(-H^l) + \gamma$$



Social Strength (EBM model)

$$s_{ij} = \alpha \cdot \exp(H_{ij}^R) + \beta \cdot \sum c_{ij}^l \times \exp(-H^l) + \gamma$$



Social Strength (EBM model)

$$s_{ij} = \alpha \cdot \exp(H_{ij}^R) + \beta \cdot \sum c_{ij}^l \times \exp(-H^l) + \gamma$$

where parameter α , β and γ can be learned from training data.



Social Strength (EBM model)

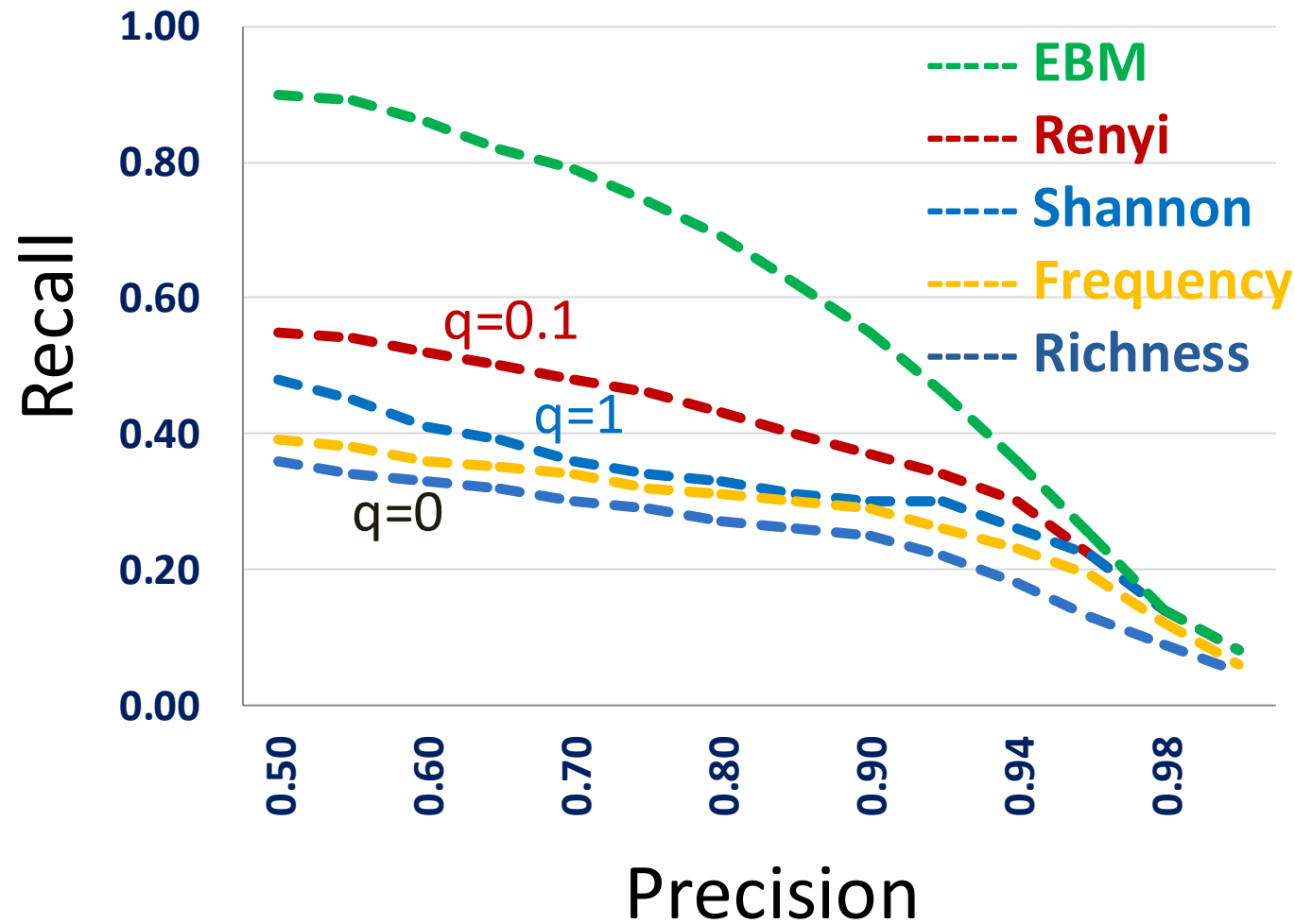
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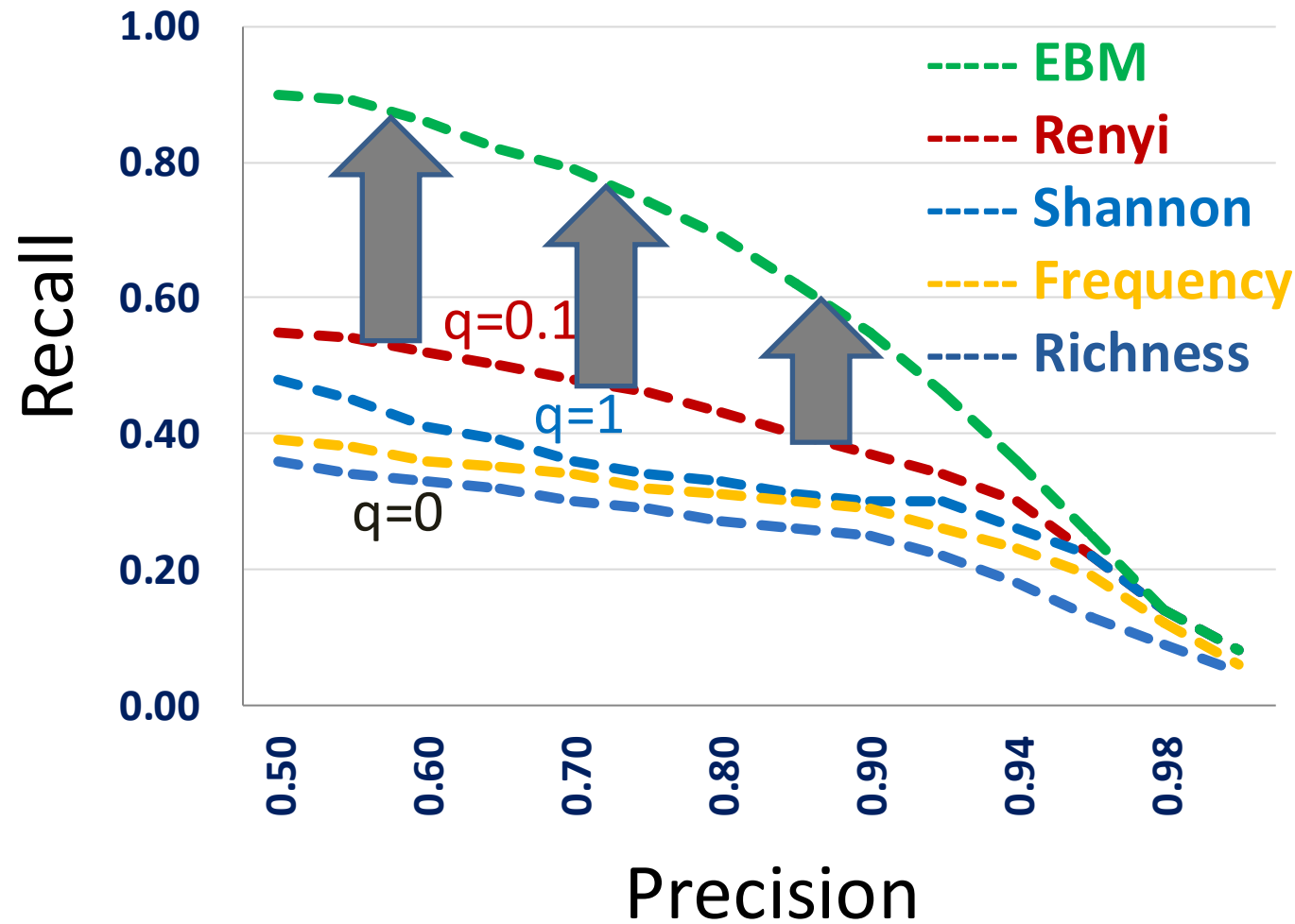
Have addressed all the challenges mentioned earlier.

- ✓ Eliminate the impact of coincidences.
- ✓ Take into account the impact of locations.
- ✓ Data Sparseness.

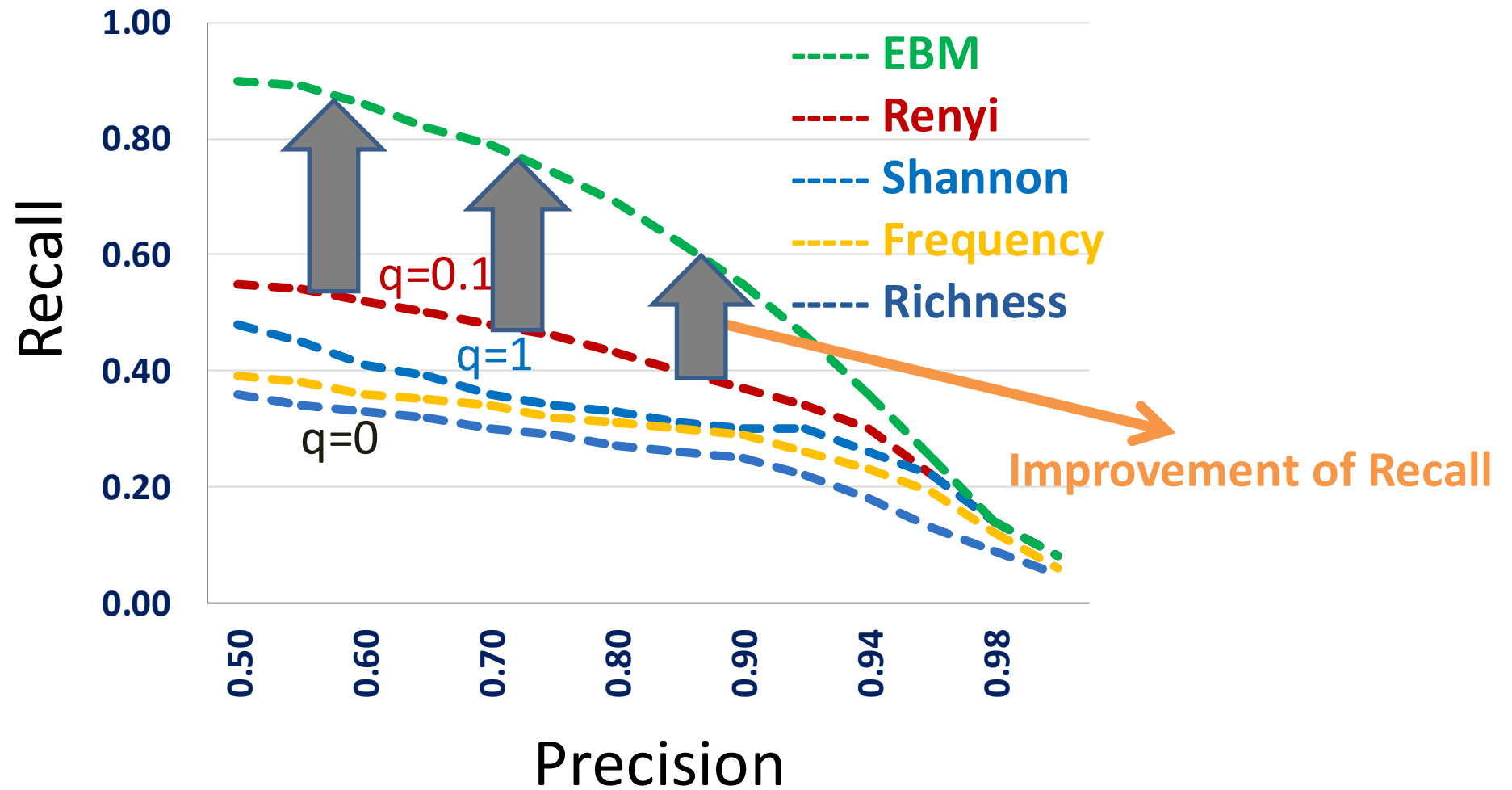
Comparison of Various Social Strength Measures



Comparison of Various Social Strength Measures



Comparison of Various Social Strength Measures



OUTLINE



GeoSocial Queries [VLDB'13]

Inferring Social from Geo [SIGMOD'13]

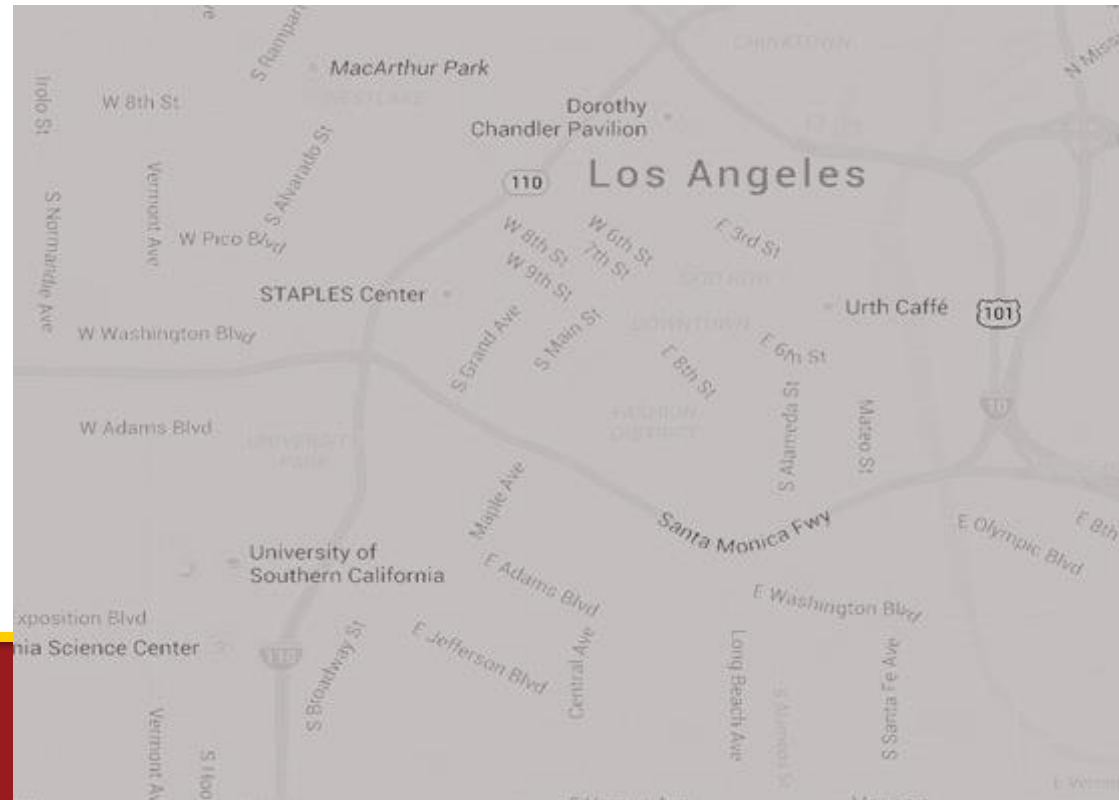
GeoSocial Recommendation [SIGMOD'15]

Future [SIGSPATIAL'15]

GeoSocial Recommendation [SIGMOD'15]



GeoSocial Recommendation [SIGMOD'15]

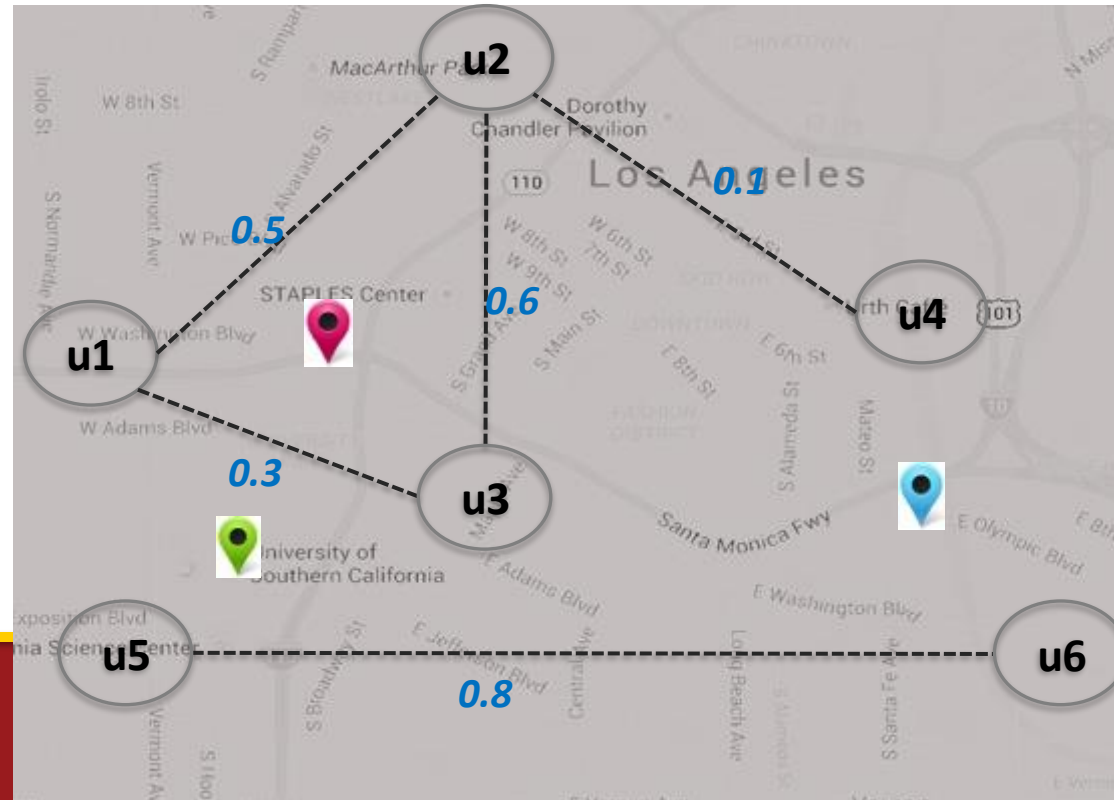


GeoSocial Recommendation [SIGMOD'15]

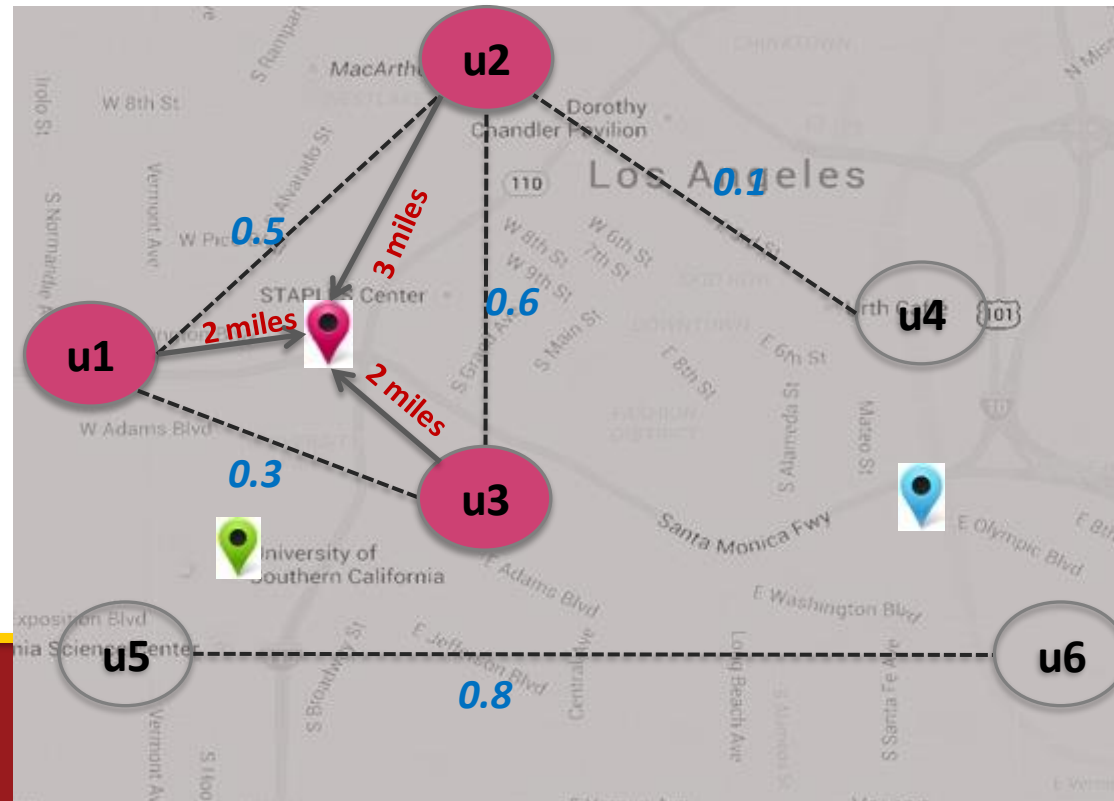




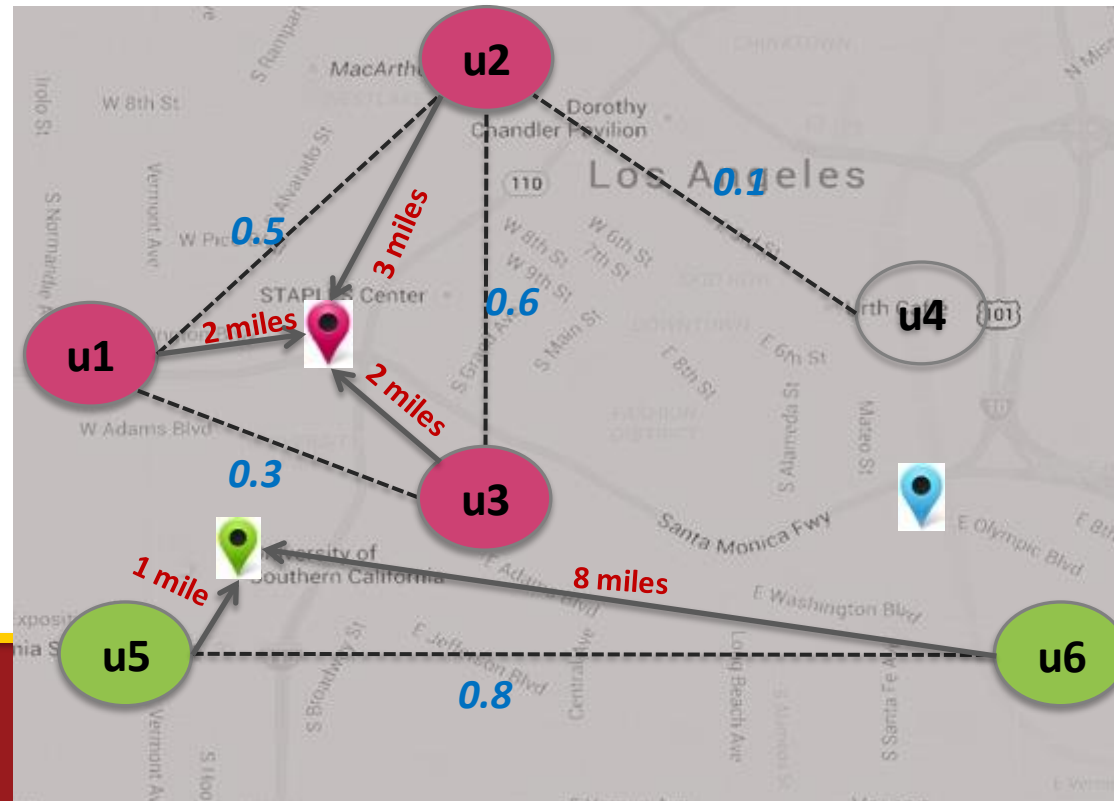
GeoSocial Recommendation [SIGMOD'15]



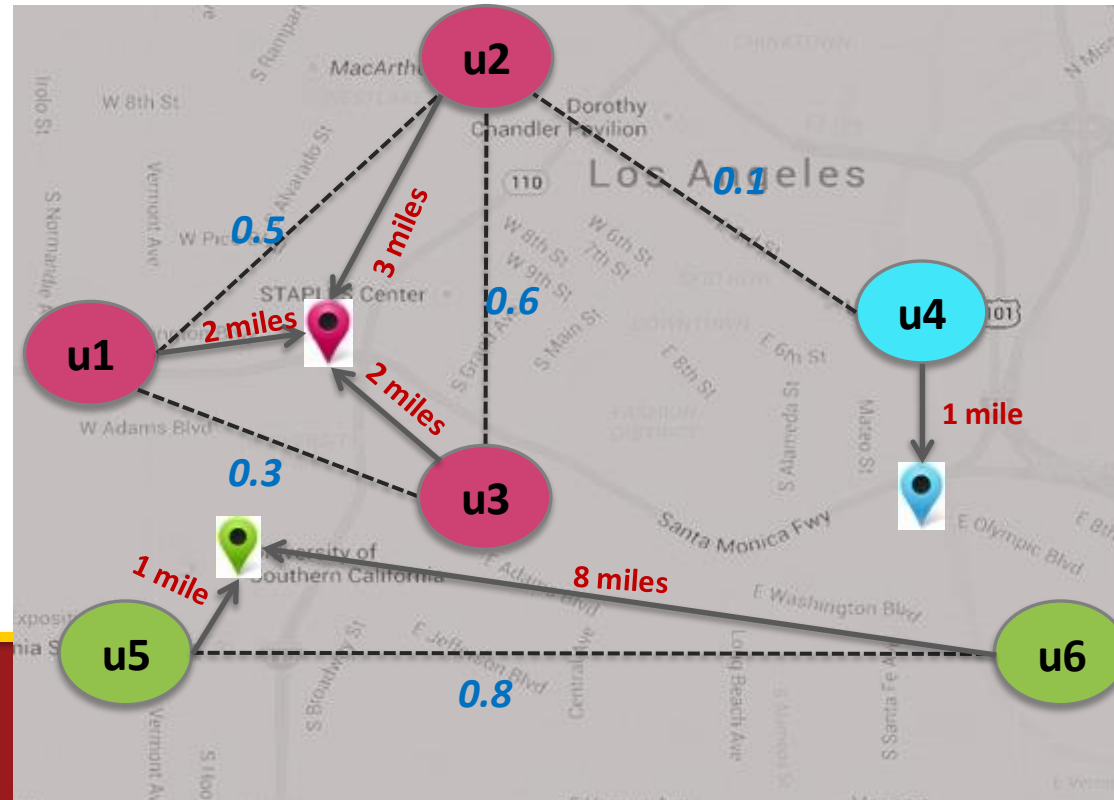
GeoSocial Recommendation [SIGMOD'15]



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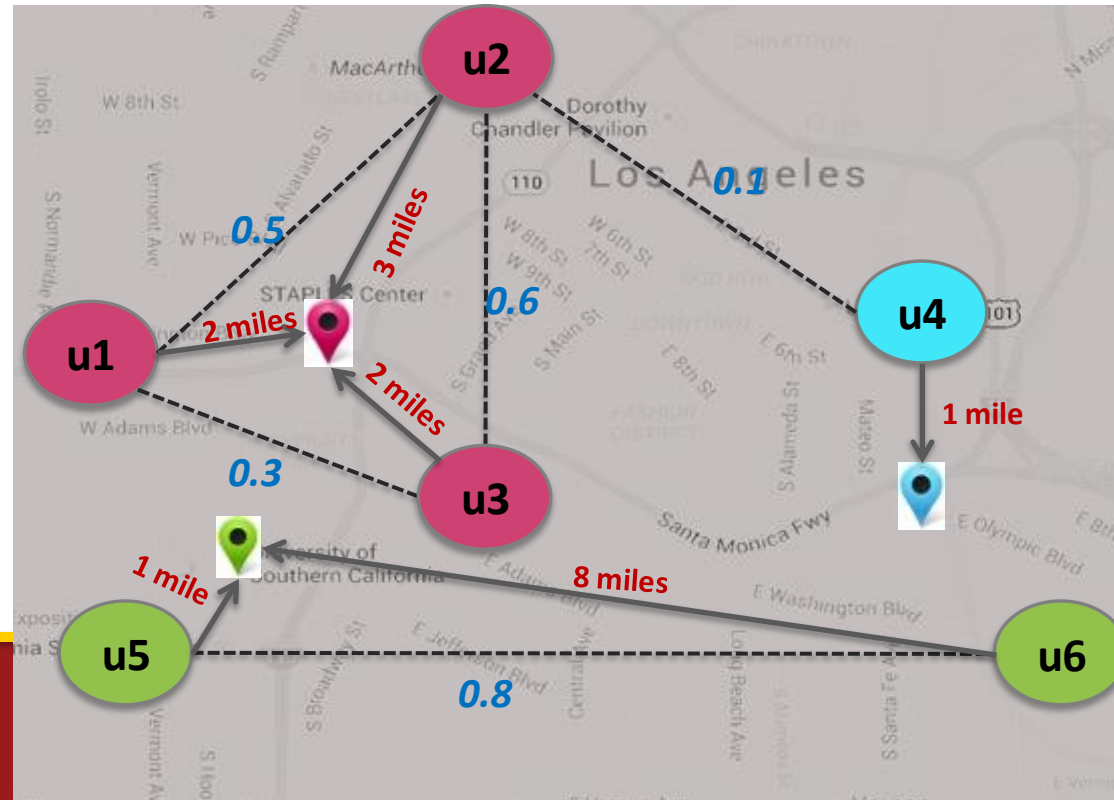


GeoSocial Recommendation [SIGMOD'15]



Real-Time Multi-Criteria Social Graph Partitioning

$$RMGP(G, P, \alpha) = \alpha \cdot \sum_{v \in V} c(v, s_v) + (1 - \alpha) \cdot \sum_{\substack{e=(v,f) \in E \wedge \\ s_v \neq s_f}} w_e \quad (1)$$



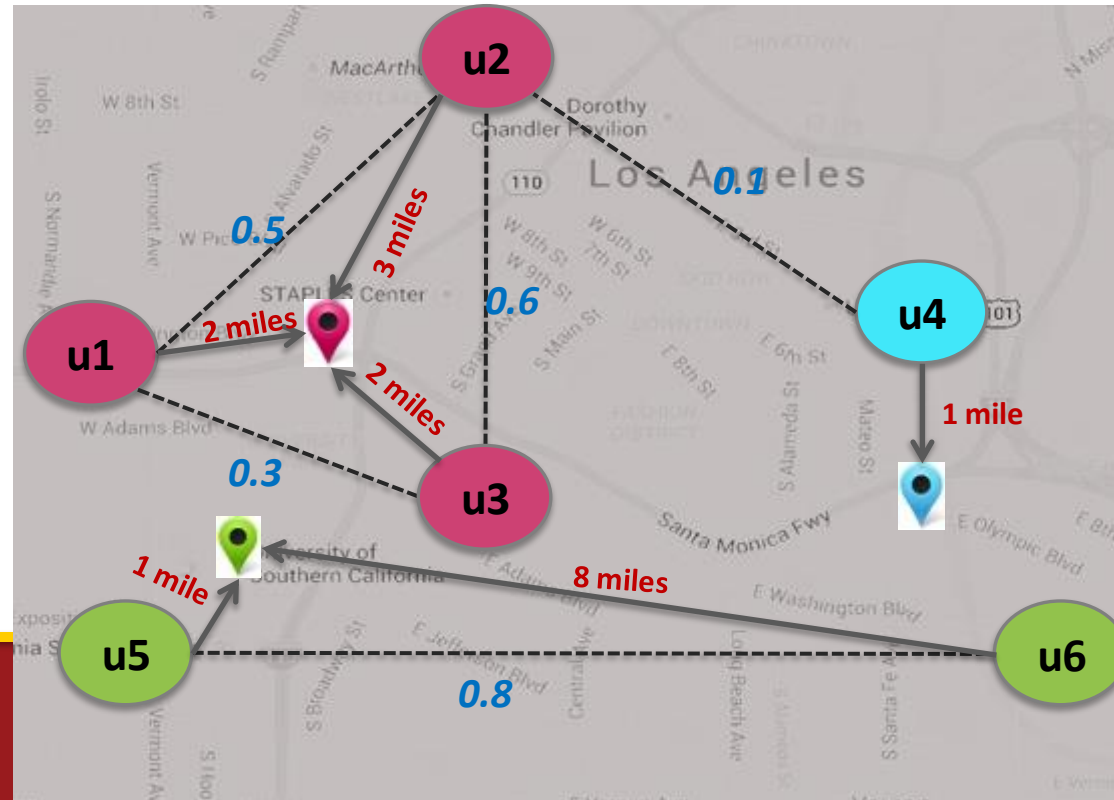
GeoSocial Recommendation [SIGMOD'15]



Real-Time Multi-Criteria Social Graph Partitioning

$$RMGP(G, P, \alpha) = \underbrace{\alpha \cdot \sum_{v \in V} c(v, s_v)}_{\text{Travel Cost}} + (1 - \alpha) \cdot \sum_{\substack{e=(v,f) \in E \wedge \\ s_v \neq s_f}} w_e \quad (1)$$

Travel Cost



GeoSocial Recommendation [SIGMOD'15]

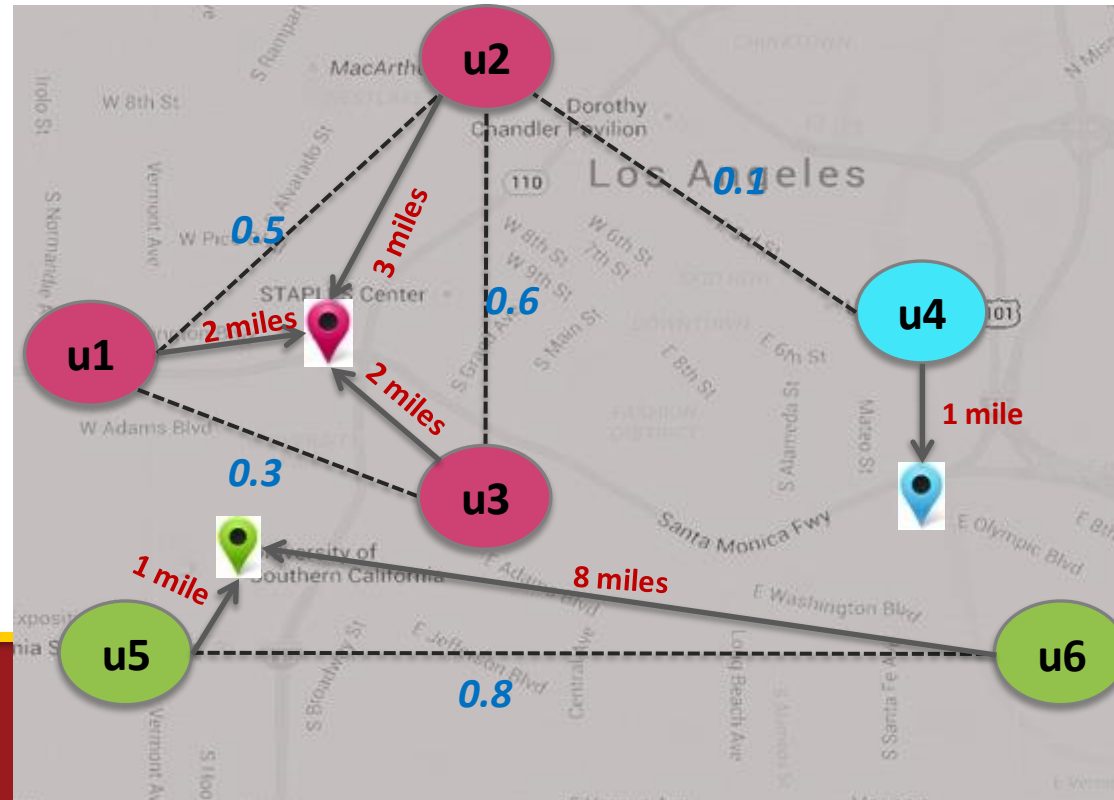


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Travel Cost

Loss of Social Connectivity



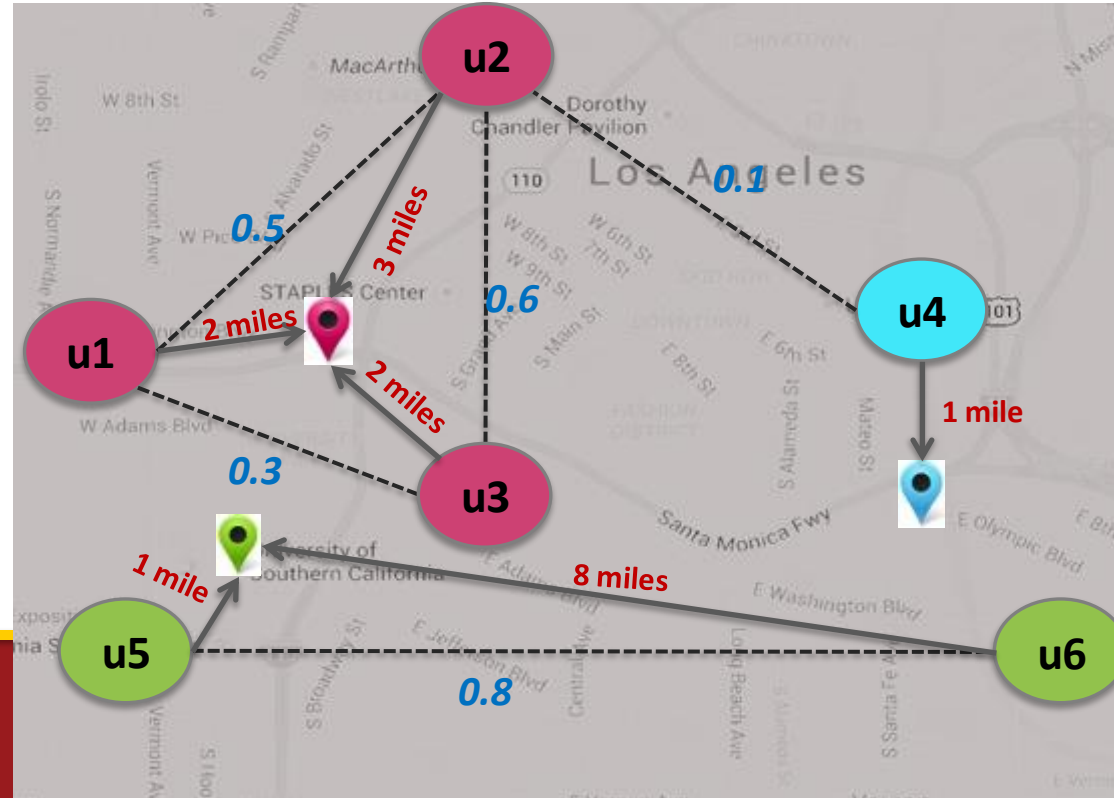
GeoSocial Recommendation [SIGMOD'15]



Real-Time Multi-Criteria Social Graph Partitioning

$$RMGP(G, P, \alpha) = \underbrace{\alpha \cdot \sum_{v \in V} c(v, s_v)}_{\text{Minimizing Total Cost}} + \underbrace{(1 - \alpha) \cdot \sum_{\substack{e=(v,f) \in E \wedge \\ s_v \neq s_f}} w_e}_{\text{Loss of Social Connectivity}} \quad (1)$$

Travel Cost



Related Work



- Graph Partitioning
 - Attribute-based [J. Sun et al., SIGKDD '07]
 - Connectivity-based [J. Shi et al., TPAMI '00], [M. E. Newman et al., Physical Review '04]
 - Attribute & Connectivity-based [Y. van Gennip et al. SIAM JAP '13]
- Uniform Metric Labeling: Same objective function as RMGP, but studied only in theory. Solutions:
 - Linear Programming (UML_{lp}), and [J. Kleinberg et al., JACM '02]
 - Greedy (UML_{gr}). [E. C. Bracht et al., JEA '05]

GAME THEORETIC APPROACH



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Every user $v \in V$ is a greedy player, who wants to attend the event s_v that minimizes his cost:

$$c_v(s_v, \bar{s}_v) = \alpha c(v, s_v) + (1 - \alpha) \sum_{(e=(v,f) \in E) \cap (s_v \cap s_f)} w_e$$

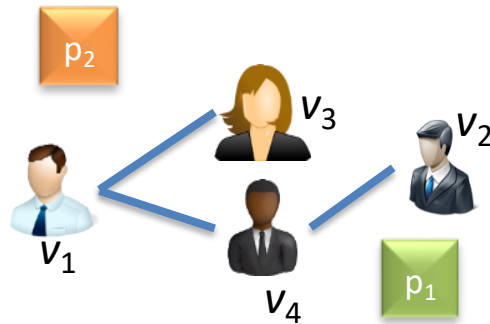
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Algorithm/Example: $\alpha=0.5$, equally weighted social edges



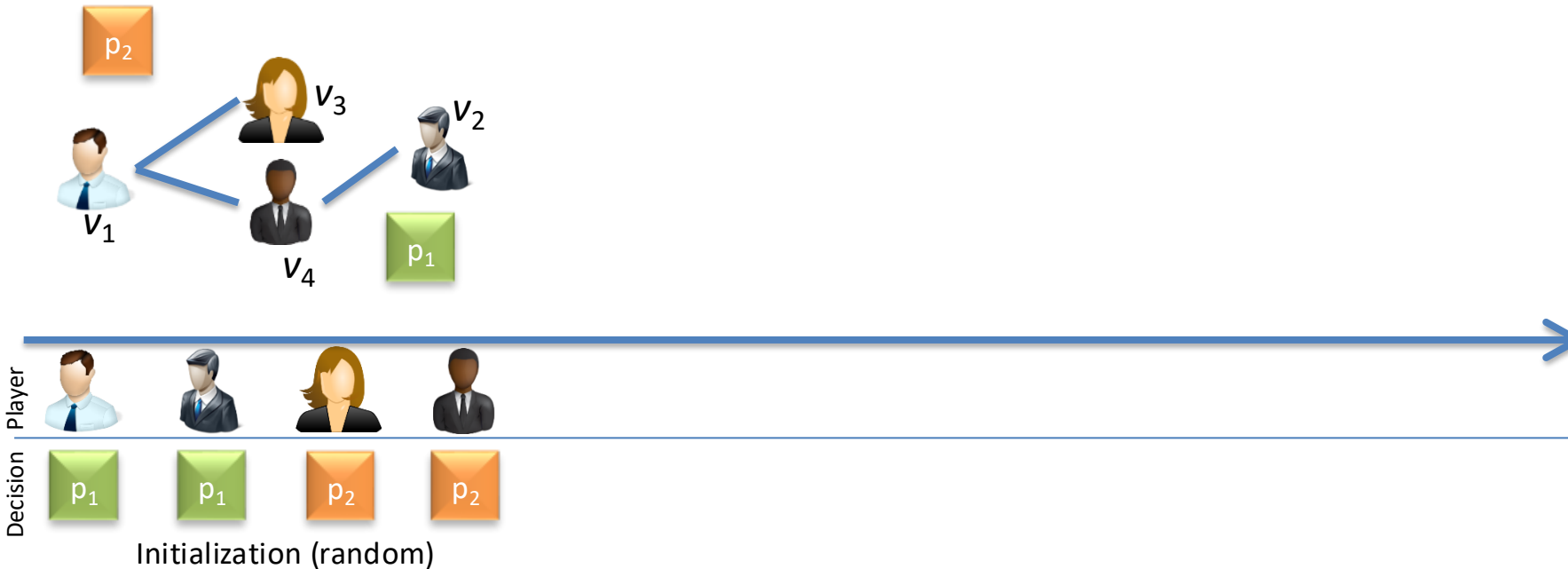
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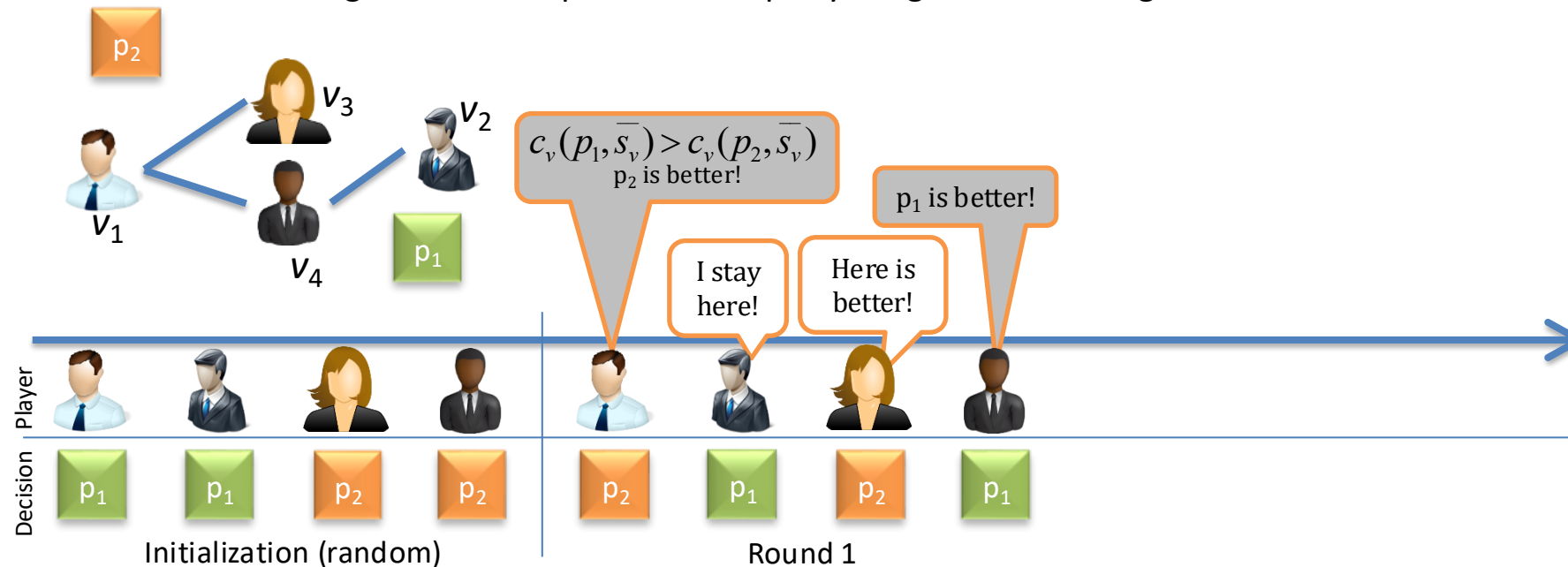
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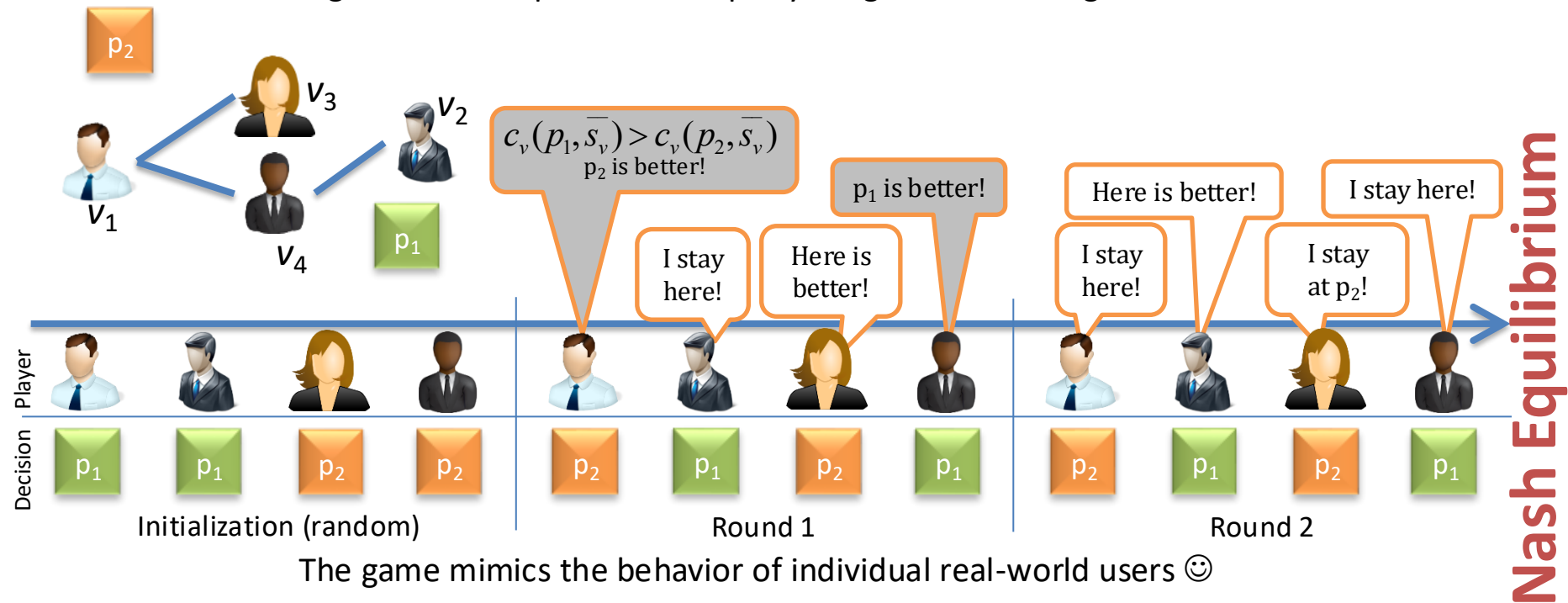
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Algorithm/Example: $\alpha=0.5$, equally weighted social edges



THEORETICAL RESULTS



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1. Our game is an *exact potential game* -> always converges.

THEORETICAL RESULTS



1. Our game is an **exact potential game** -> always converges.

- Potential function:

$$\Phi(S) = \alpha \sum_{v \in V} c(v, s_v) + (1 - \alpha) \sum_{(e=(v,f) \in E) \cap (s_v \neq s_f)} \frac{1}{2} w_e$$

- When a user v moves from s_v to s'_v then:

$$C_v(s_v, \bar{s}_v) - C_v(s'_v, \bar{s}_v) = \Phi(s_v, \bar{s}_v) - \Phi(s'_v, \bar{s}_v)$$

[D. Monderer et al., Games and economic behavior, 1996]

THEORETICAL RESULTS



1. Our game is an **exact potential game** -> always converges.

- Potential function:

$$\Phi(S) = \alpha \sum_{v \in V} c(v, s_v) + (1 - \alpha) \sum_{(e=(v,f) \in E) \cap (s_v \neq s_f)} \frac{1}{2} w_e$$

- When a user v moves from s_v to s'_v then:

$$C_v(s_v, \bar{s}_v) - C_v(s'_v, \bar{s}_v) = \Phi(s_v, \bar{s}_v) - \Phi(s'_v, \bar{s}_v)$$

[D. Monderer et al., Games and economic behavior, 1996]

2. Price of anarchy is upper-bounded:

$$\frac{\text{cost of worst equilibrium}}{\text{global optimum}} \leq 1 + \frac{(1 - \alpha)}{\alpha} \frac{\text{deg}_{avg} w_{avg}}{2 c_{avg}^*}$$

deg_{avg} average degree

OUTLINE



GeoSocial Queries [VLDB'13]

Inferring Social from Geo [SIGMOD'13]

GeoSocial Recommendation [SIGMOD'15]

Future [SIGSPATIAL'15]

PLACE: Privacy-preserving Location Analytics and Computation Environment



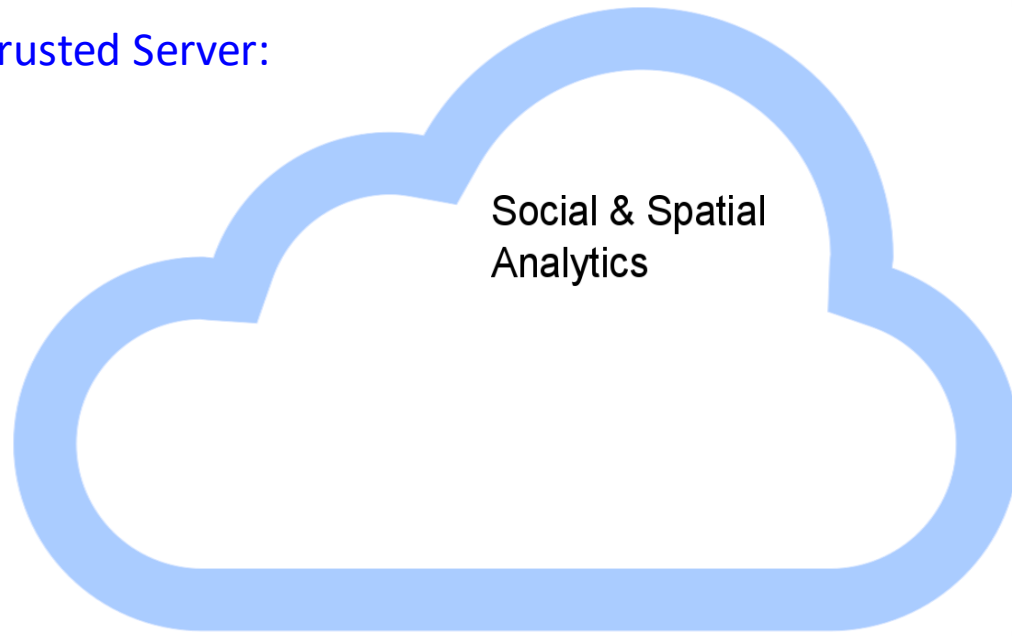
DATA CONSUMERS

from research, urban policy,
law enforcement, health policy



Untrusted Server:

Social & Spatial
Analytics



DATA PROVIDERS

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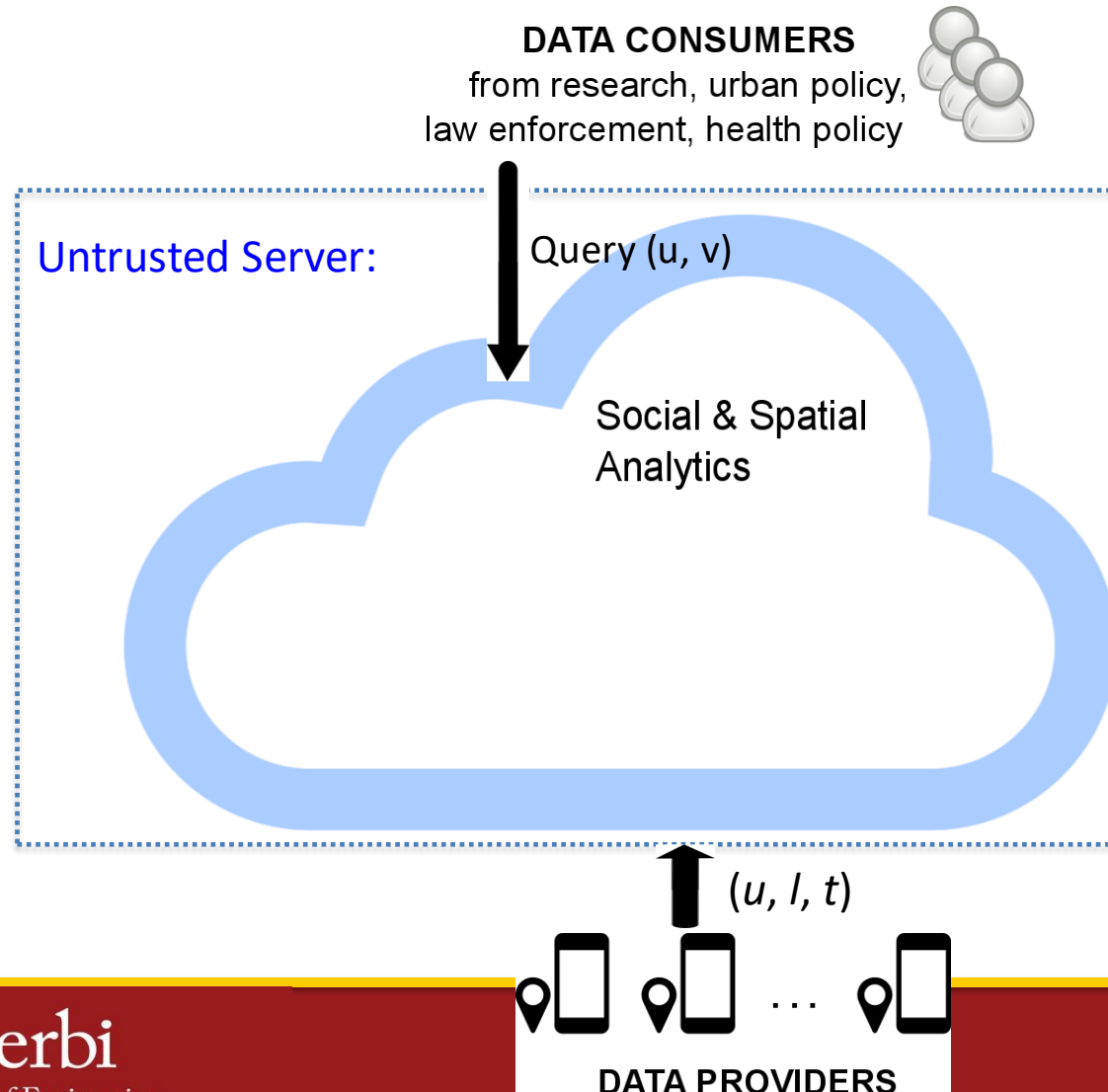
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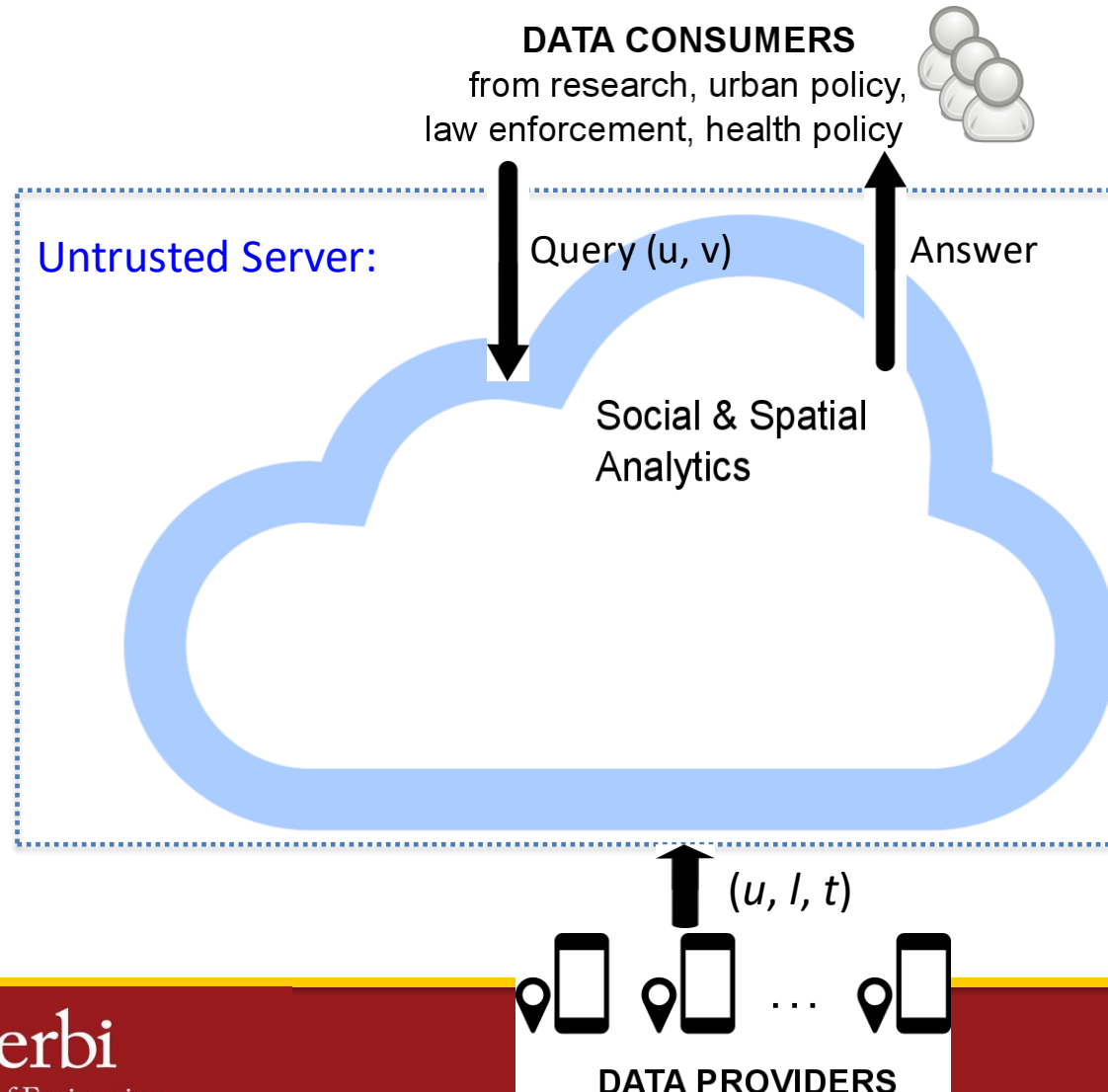


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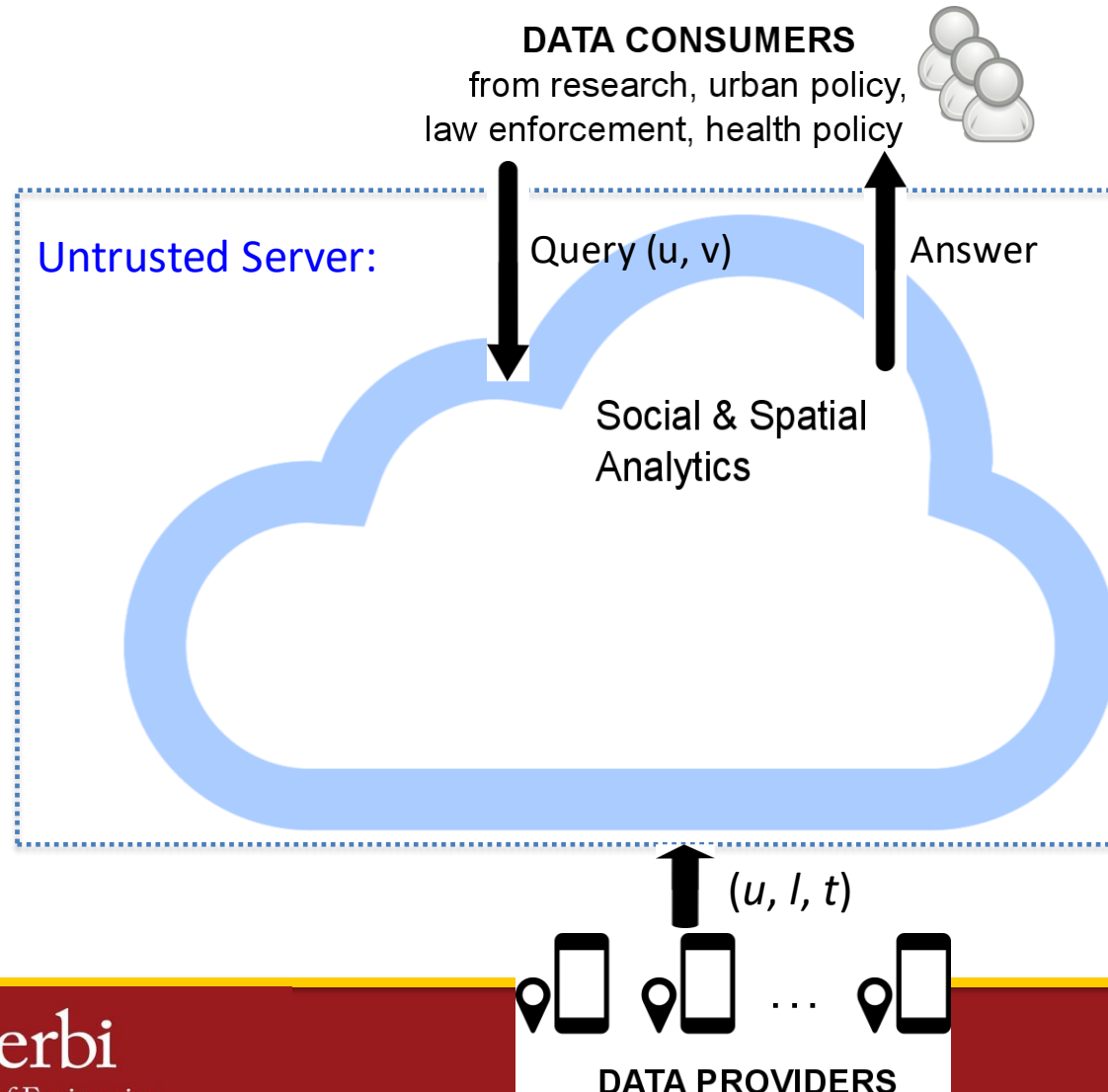
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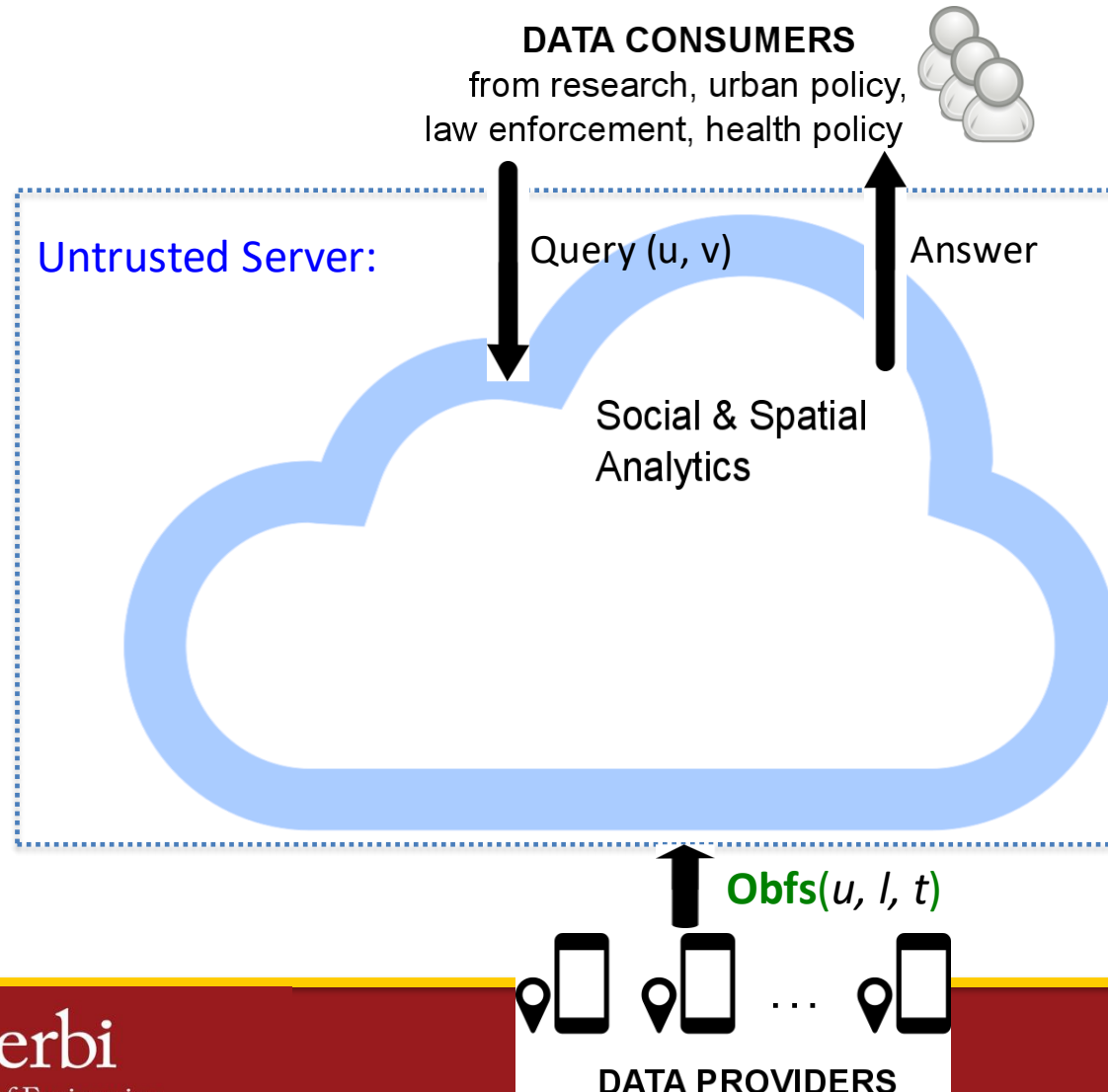
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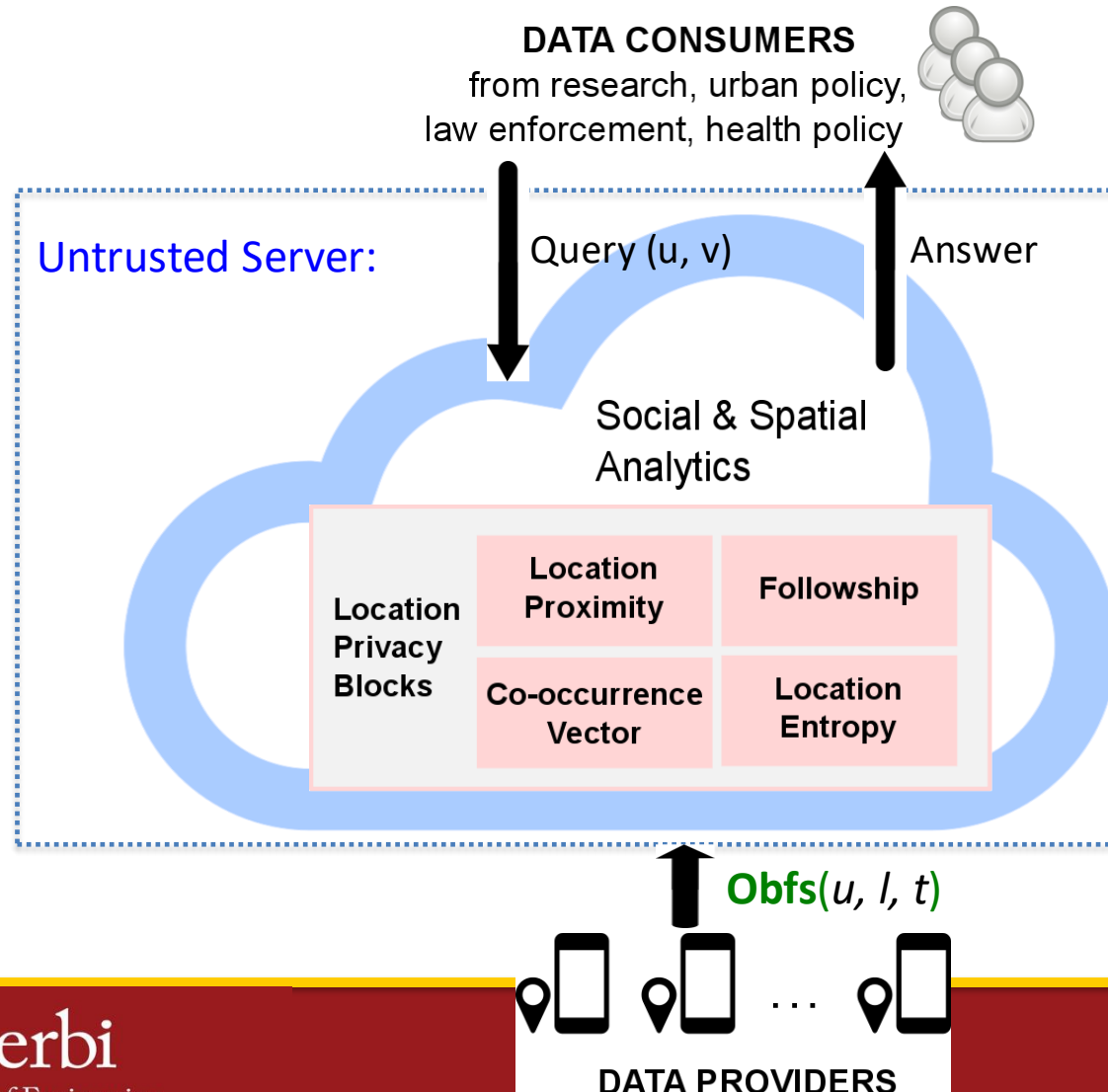
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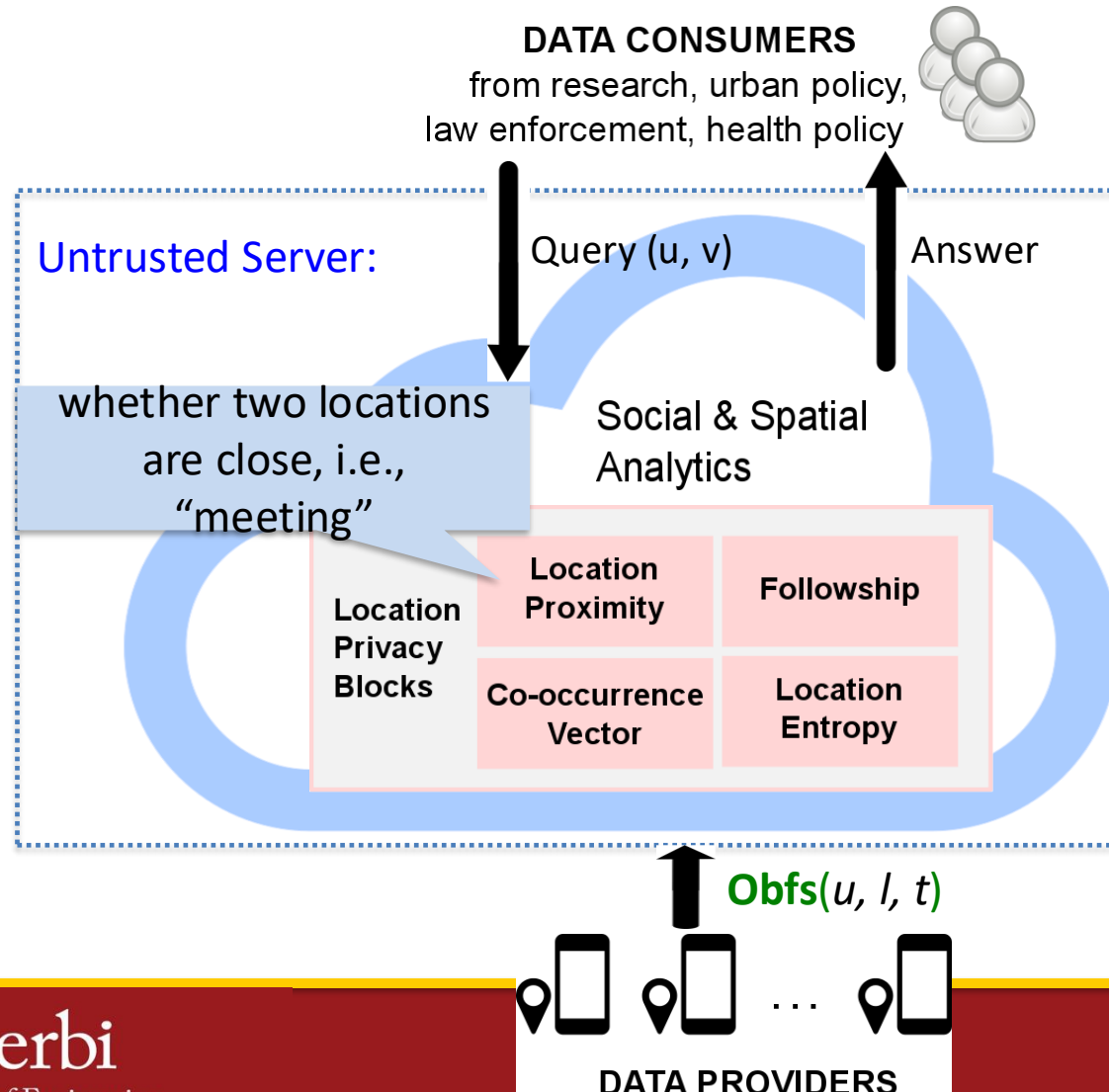
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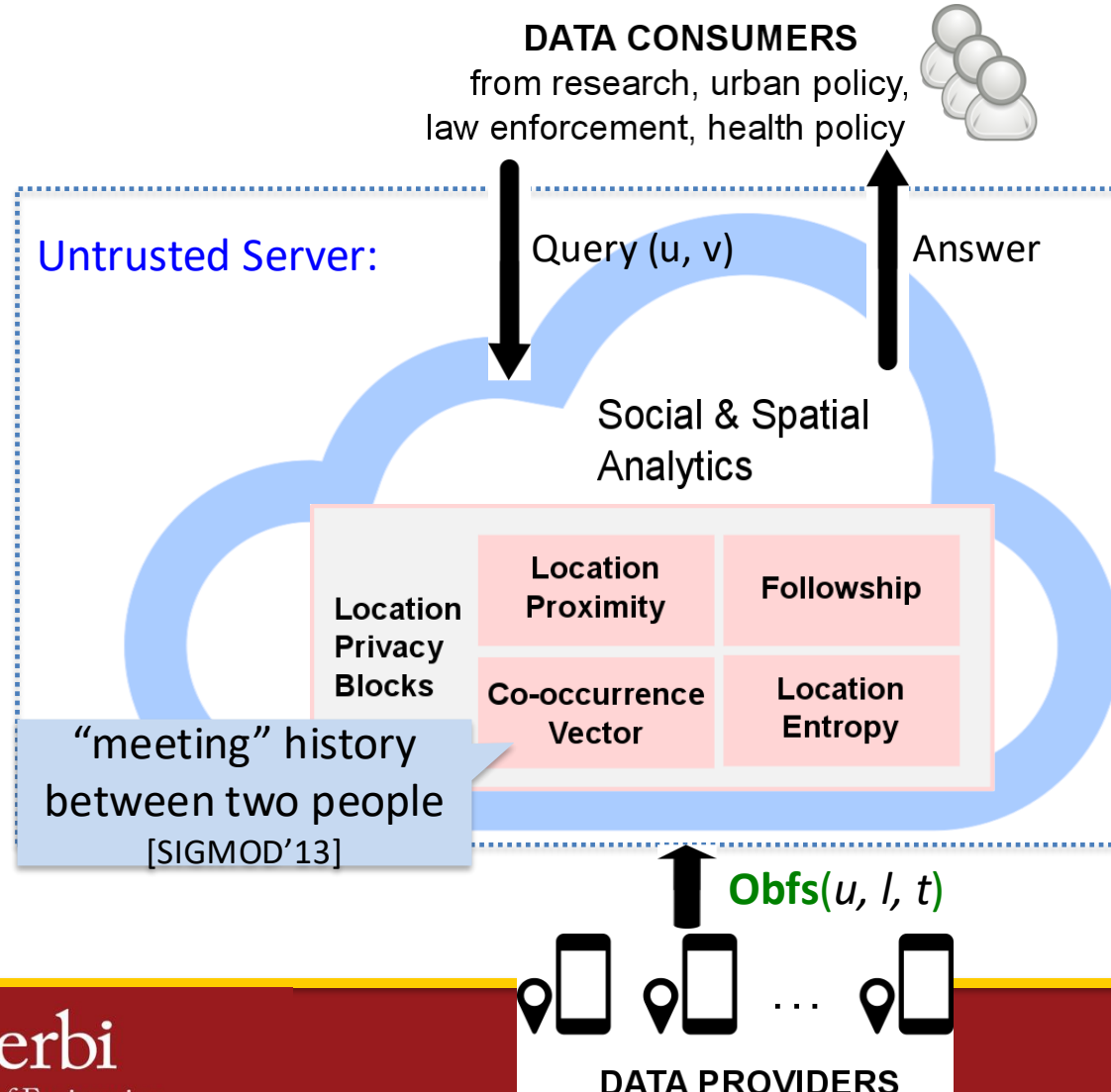
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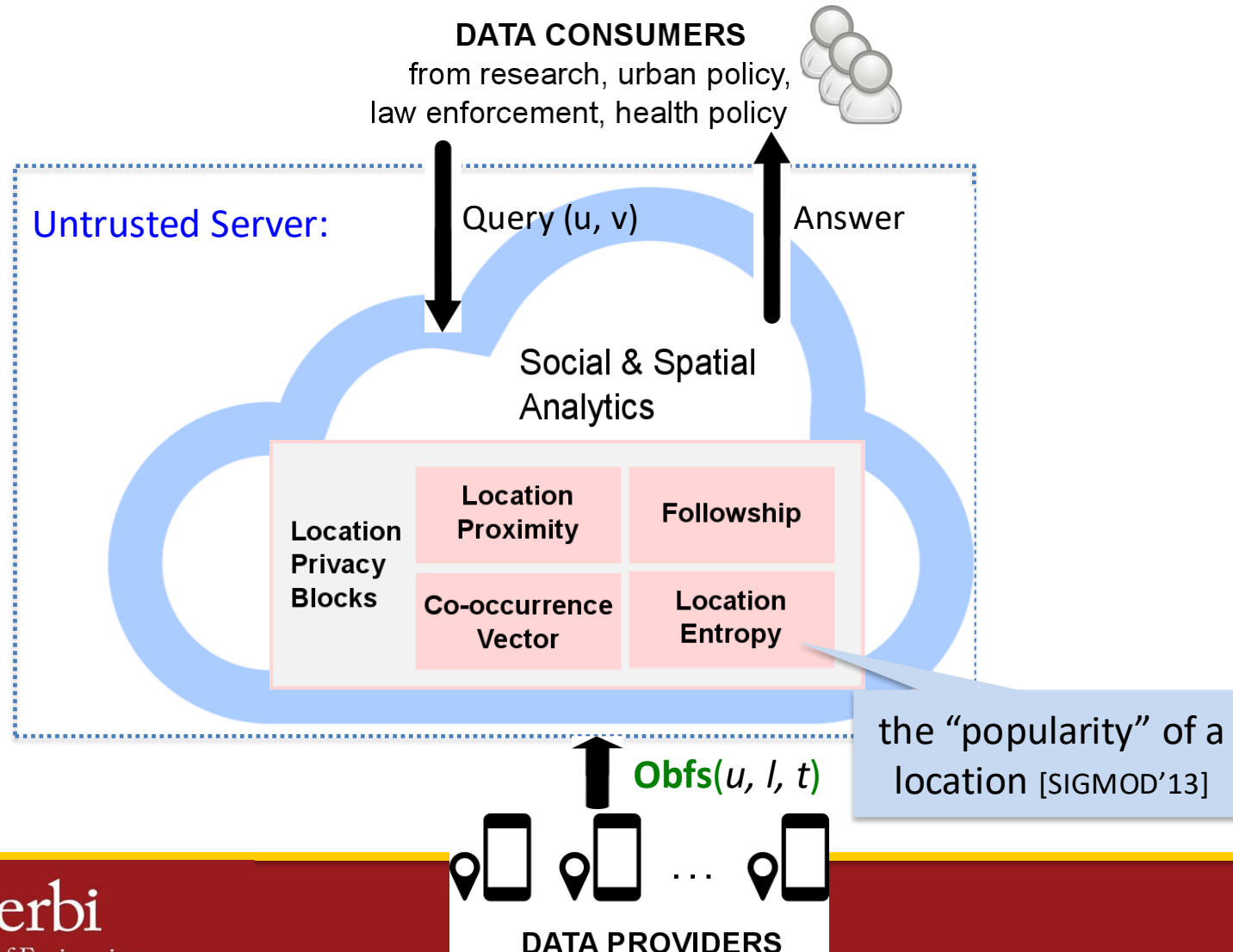
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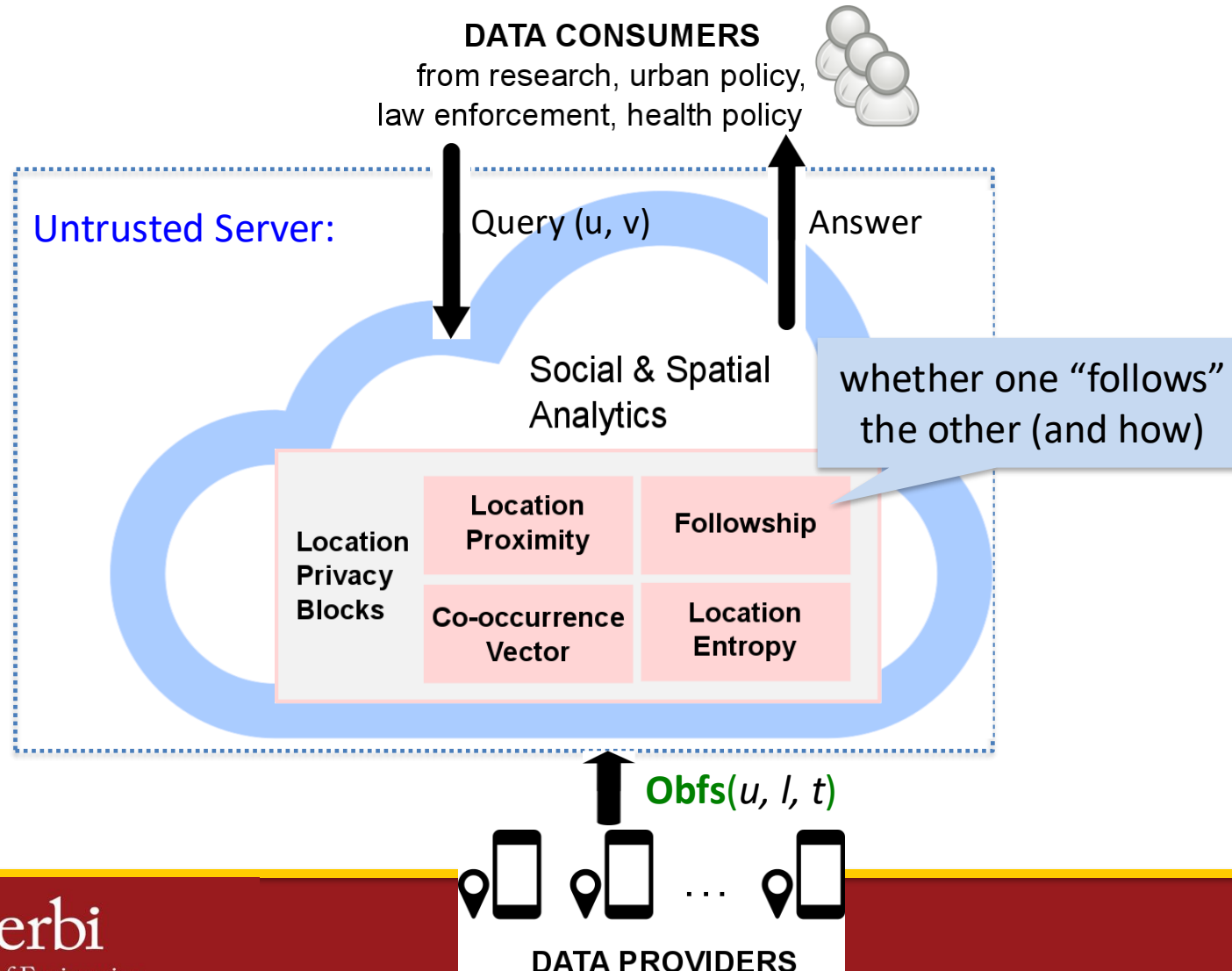
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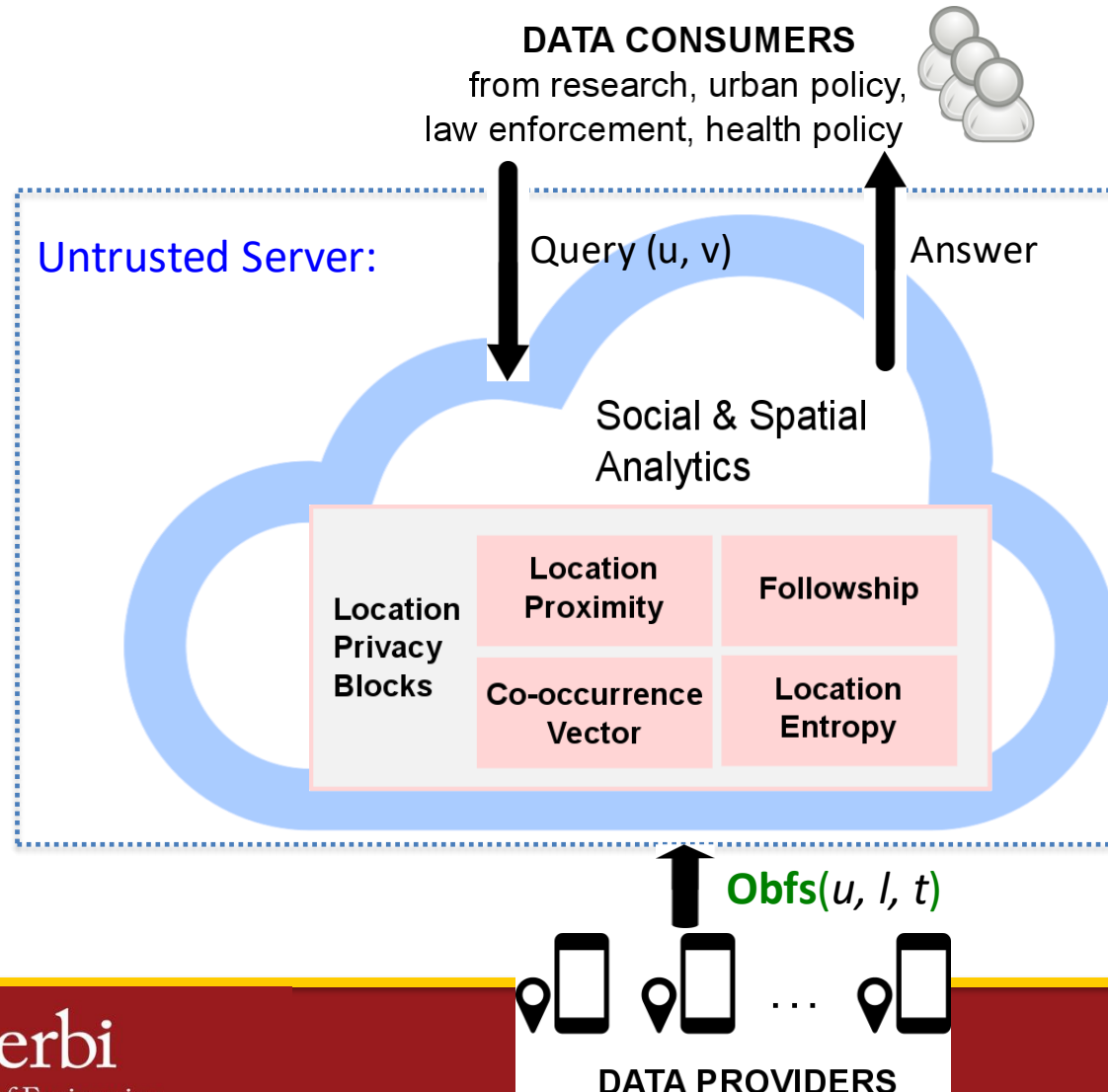
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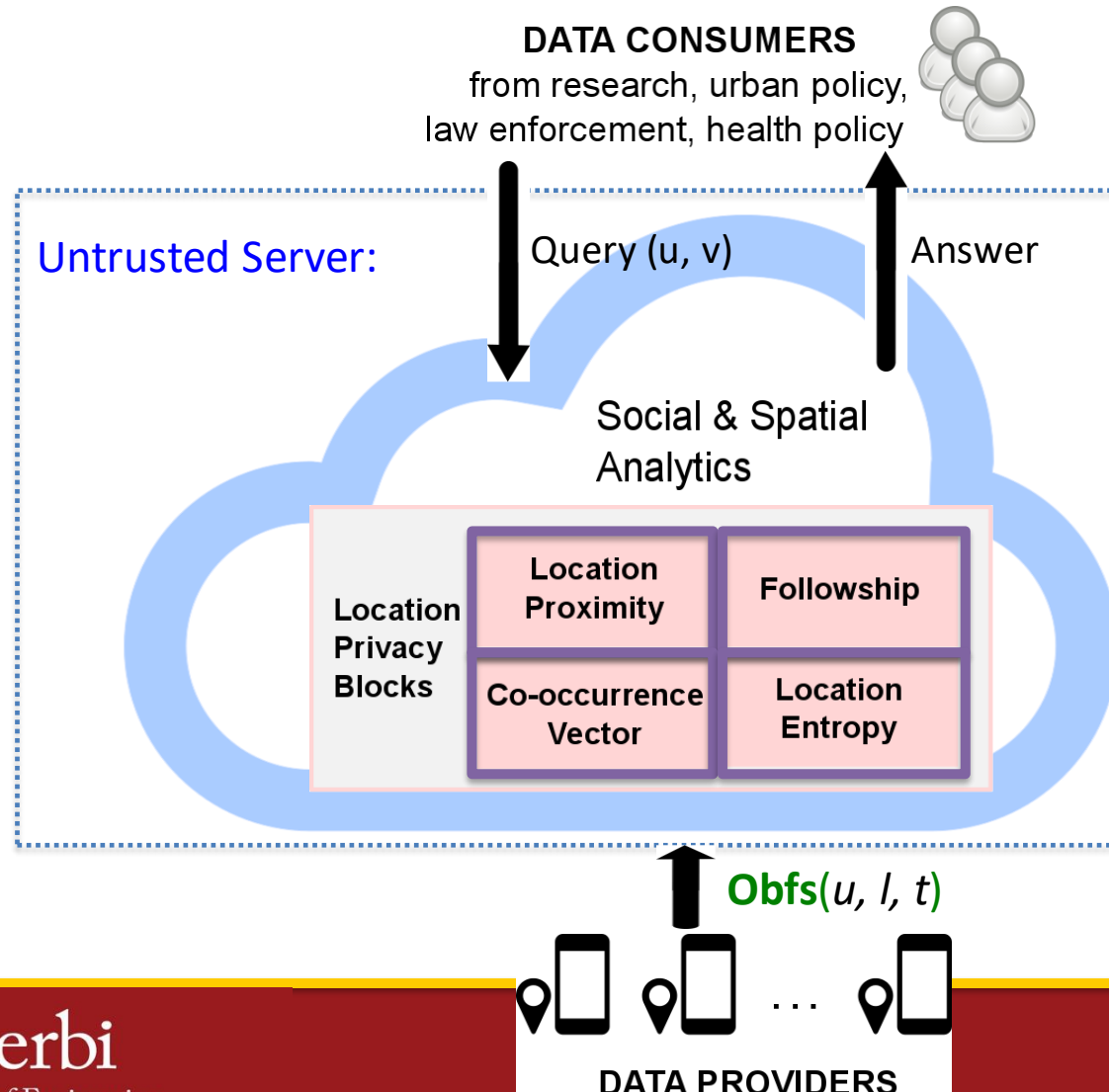
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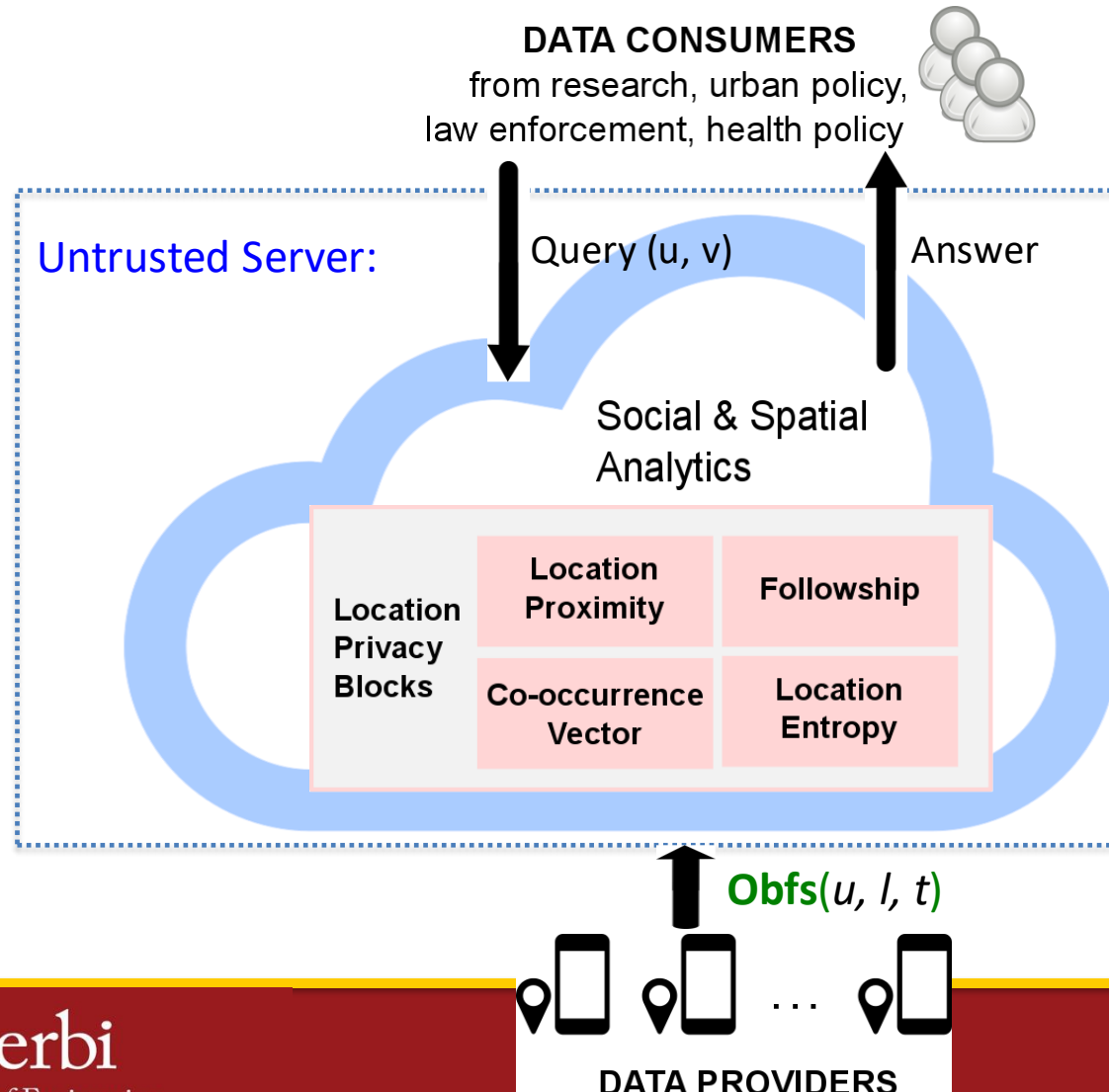
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Location obfuscation:

- Deterministic & Probabilistic Encryption [CNS'15]
- Efficiency challenge

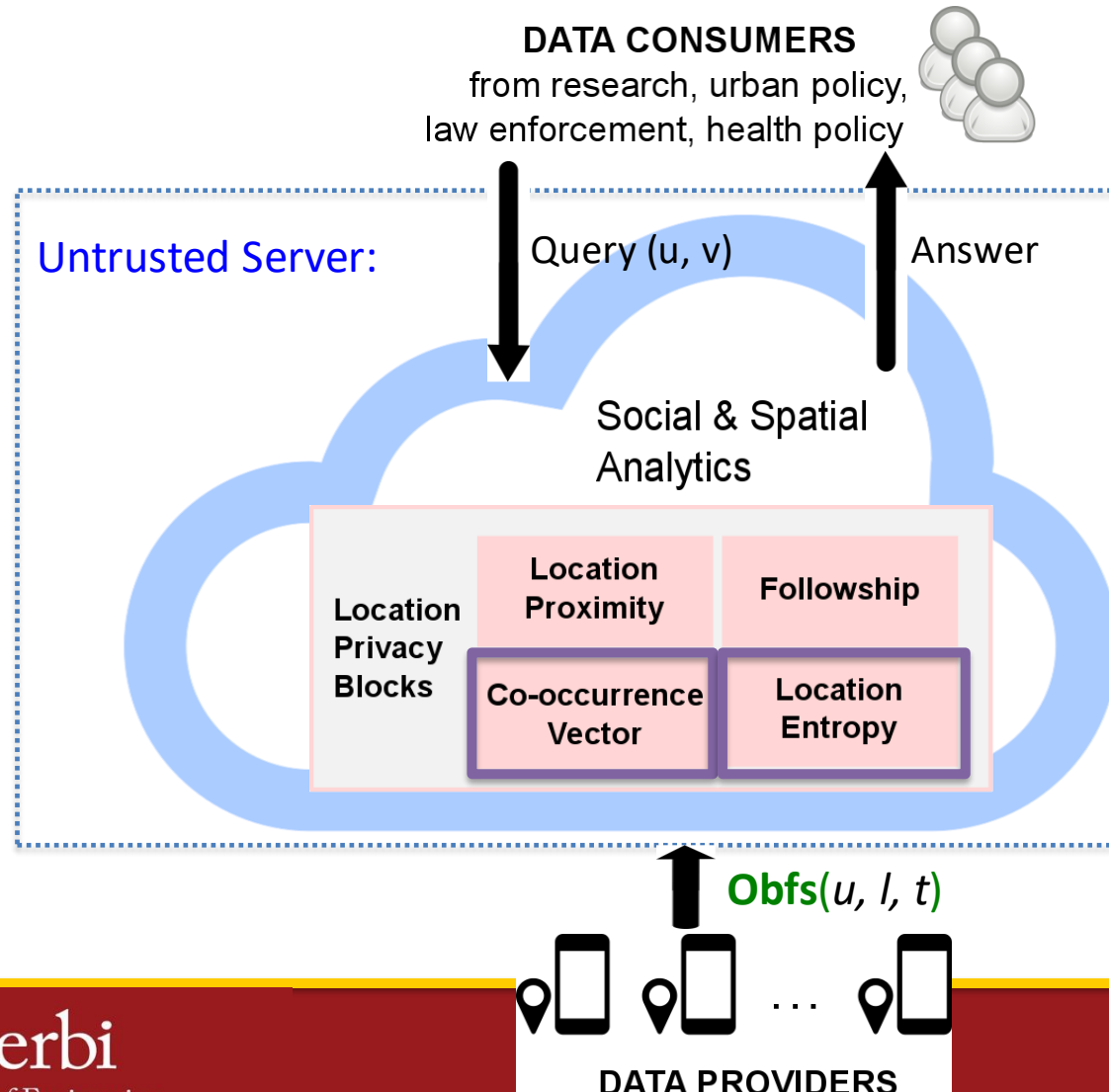
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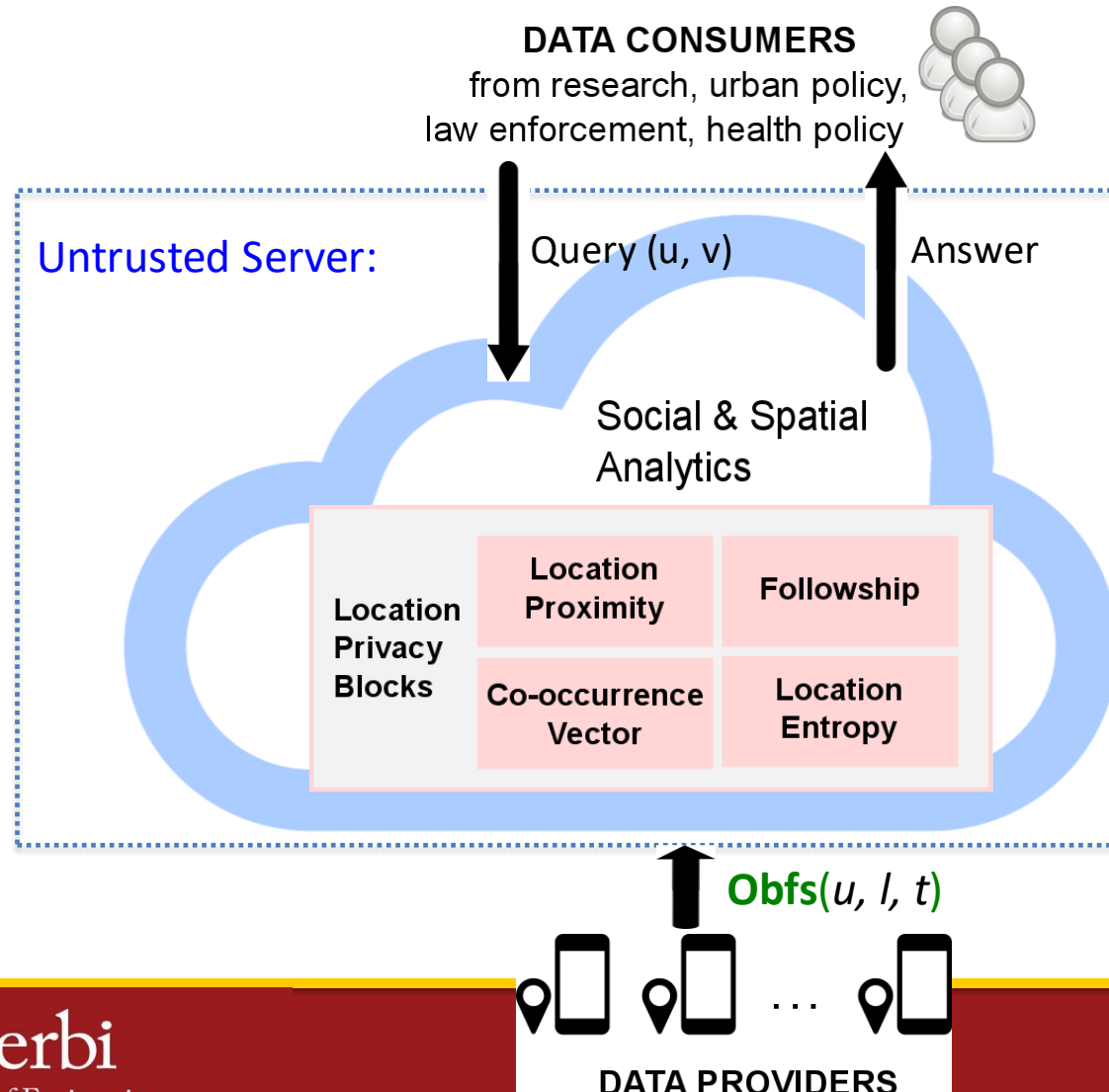
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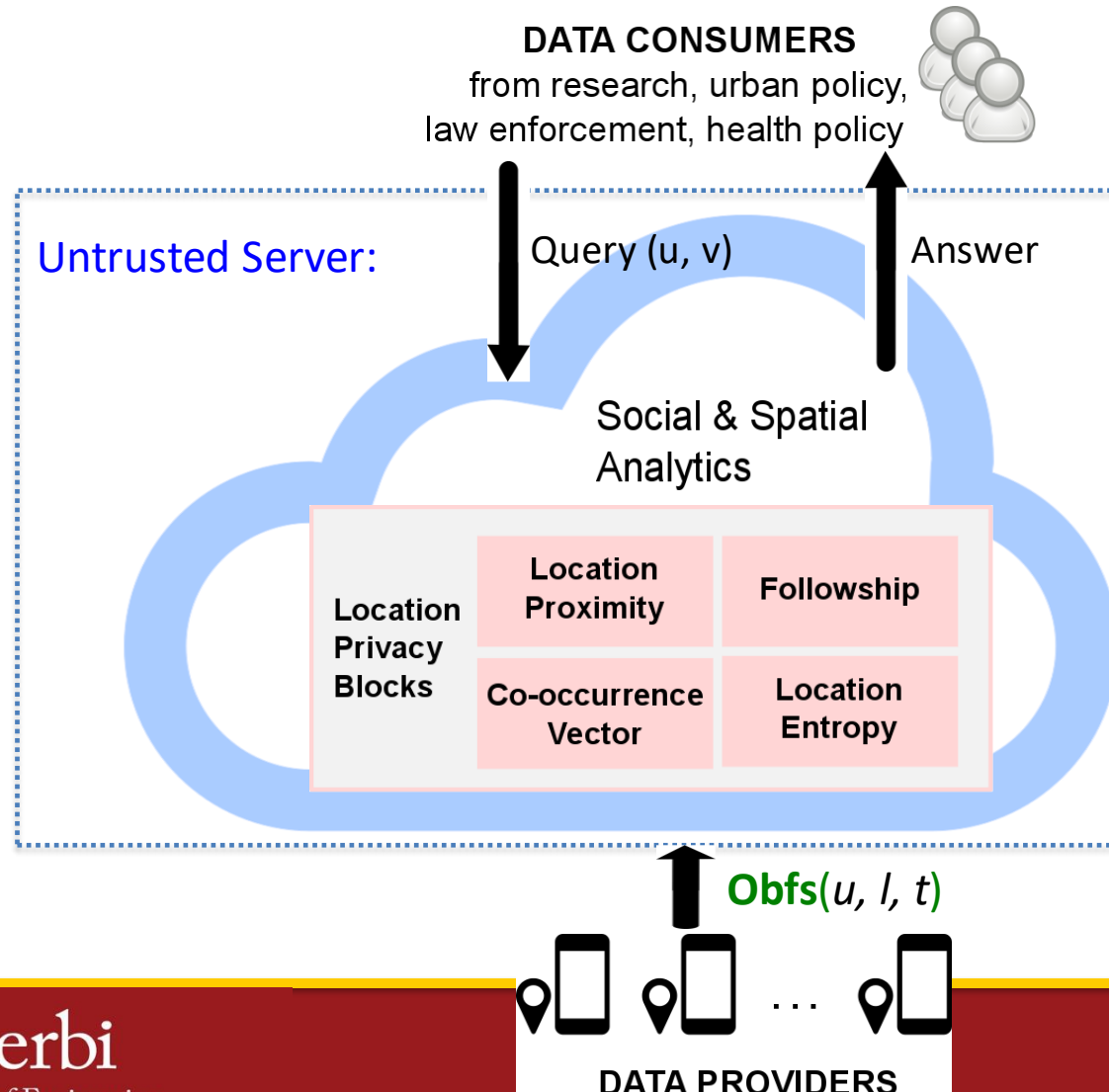
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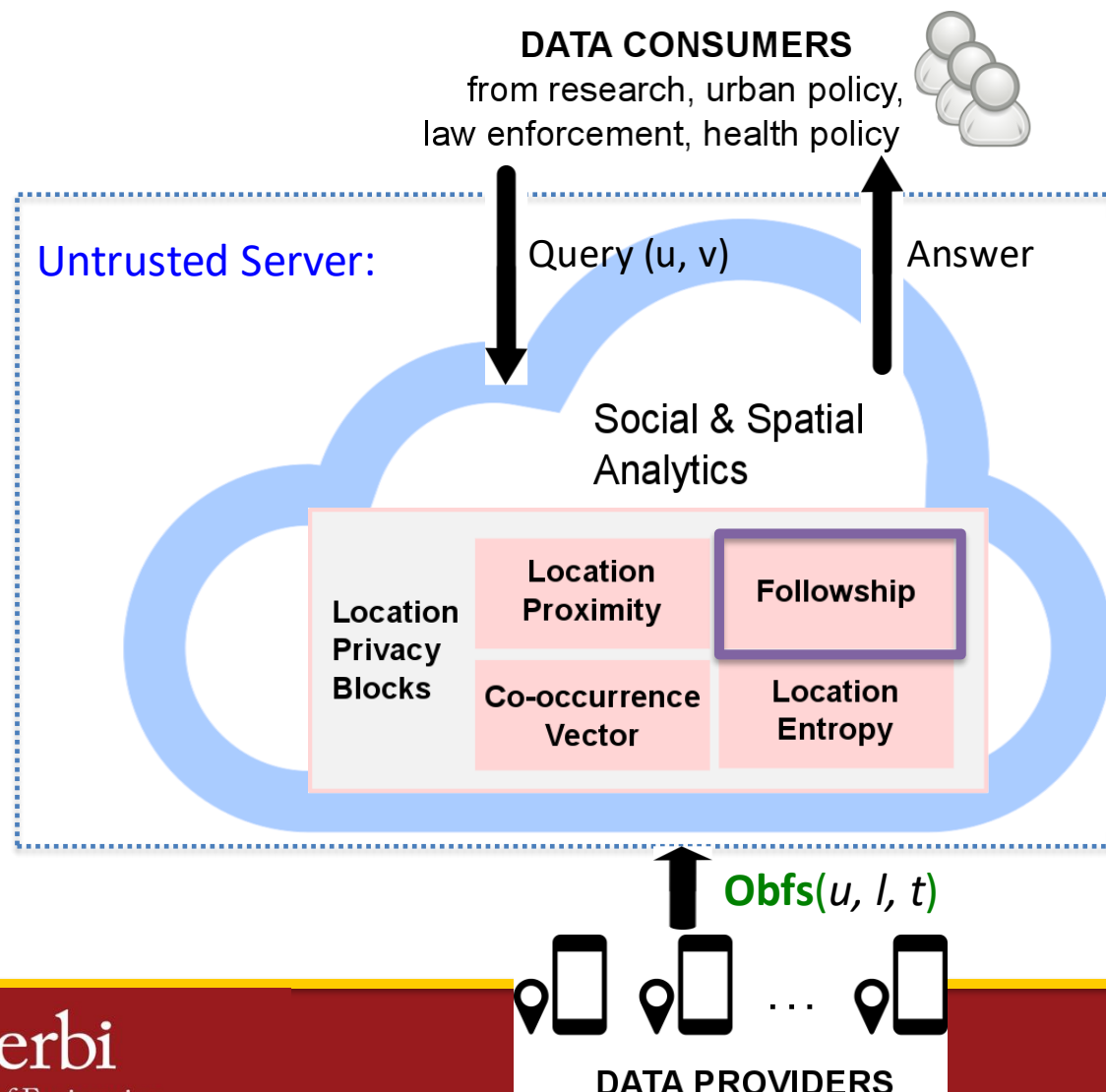
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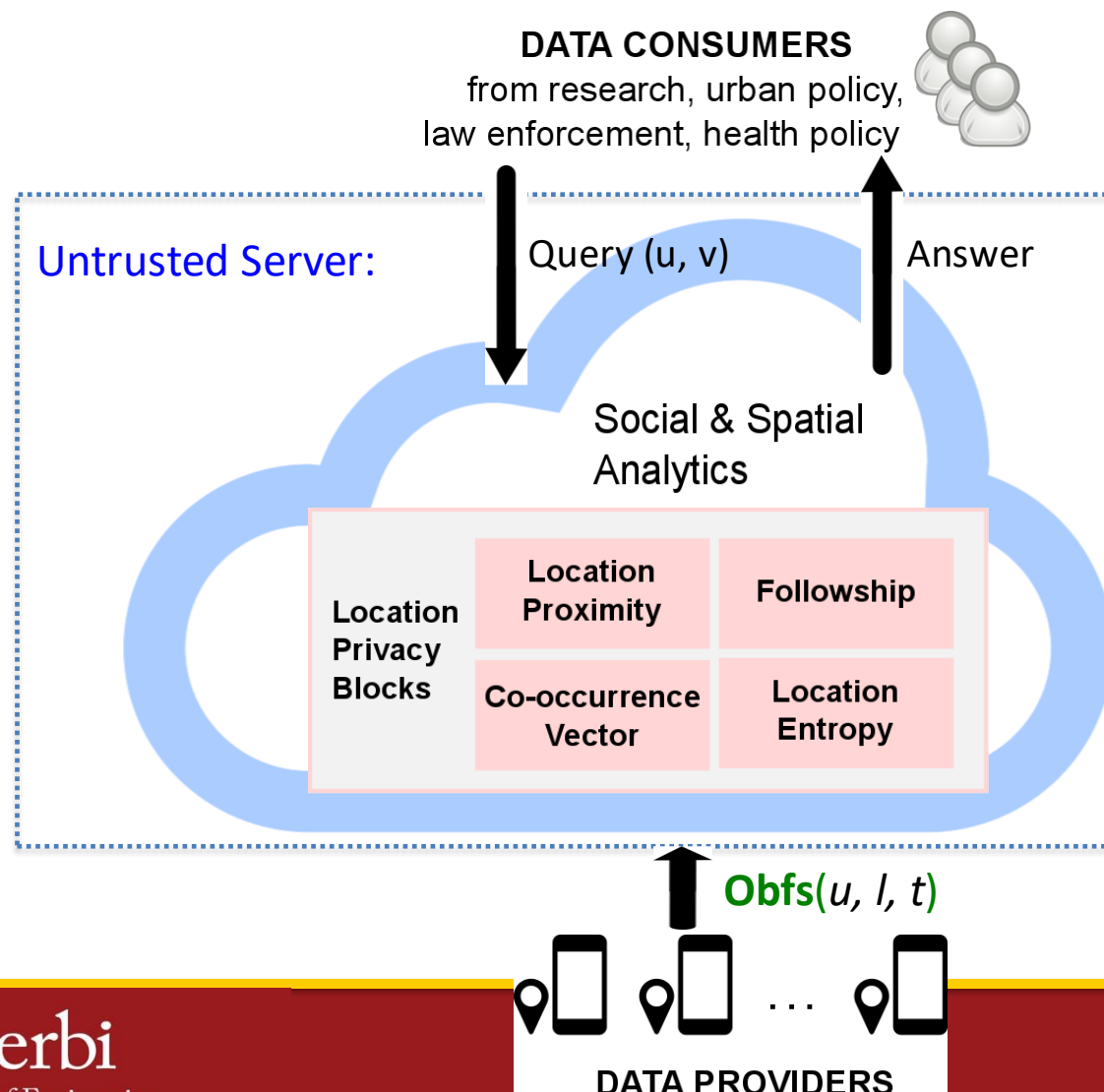
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Intermediate disclosure:

- Pseudo user_ids

Two Sides of the Coin



*Protecting against
social inferences
* But allow for LBS*

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*Protecting against
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- **[VLDB'13]** Nikos Armenatzoglou, [Stavros Papadopoulos](#), [Dimitris Papadias](#): **A General Framework for Geo-Social Query Processing**. [PVLDB 6\(10\)](#): 913-924 (2013)
- **[SIGMOD'13]** [Huy Pham](#), Cyrus Shahabi, [Yan Liu](#): **EBM: an entropy-based model to infer social strength from spatiotemporal data**. [SIGMOD Conference 2013](#): 265-276
- **[SIGMOD'15]** [Nikos Armenatzoglou](#), [Huy Pham](#), [Vasilis Ntranos](#), [Dimitris Papadias](#), Cyrus Shahabi: **Real-Time Multi-Criteria Social Graph Partitioning: A Game Theoretic Approach**. [SIGMOD Conference 2015](#): 1617-1628
- **[SIGSPATIAL'15]** Cyrus Shahabi, Liyue Fan, Luciano Nocera, Li Xiong, Ming Li: **Privacy-Preserving Inference of Social Relationships from Location Data: A Vision Paper**. ACM SIGSPATIAL Conference 2015, Seattle, Washington, Nov. 2015.